

Basic Principles of Medicine II
Module : Nervous System and Behavioral
Sciences

Lecture No: NB 03

Organisation of the Nervous System

Neuroscience Faculty
St. George's University

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Note: The lecture slides that are available for download and print contain complementary slides that are not presented during the lecture.

These slides have a blue background and contain additional explanation of the material discussed during the lecture.

Objectives

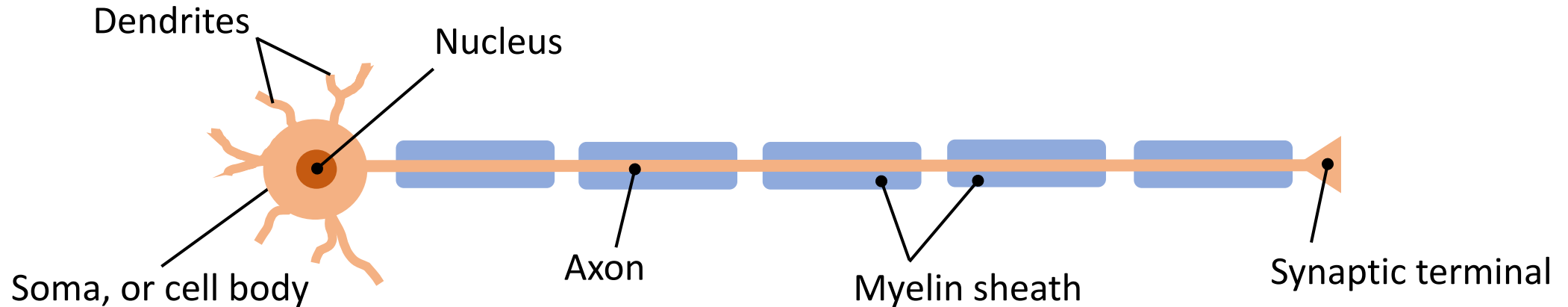
A student should be able to:

- **SOM.MK.I.BPM2.1.NB.1.NSCI.0022** Identify, differentiate and functionally characterize cerebral gray matter (nuclei) and white matter (tracts)
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0005** Describe communications between the central and peripheral nervous systems
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0552** Localize and functionally characterize the cranial nerves from surface views
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0004** Identify five major segmented parts of the spinal cord
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0006** Describe the transitionary zone between central and peripheral nervous systems
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0553** Describe the basic principles of somatic motor and sensory pathways
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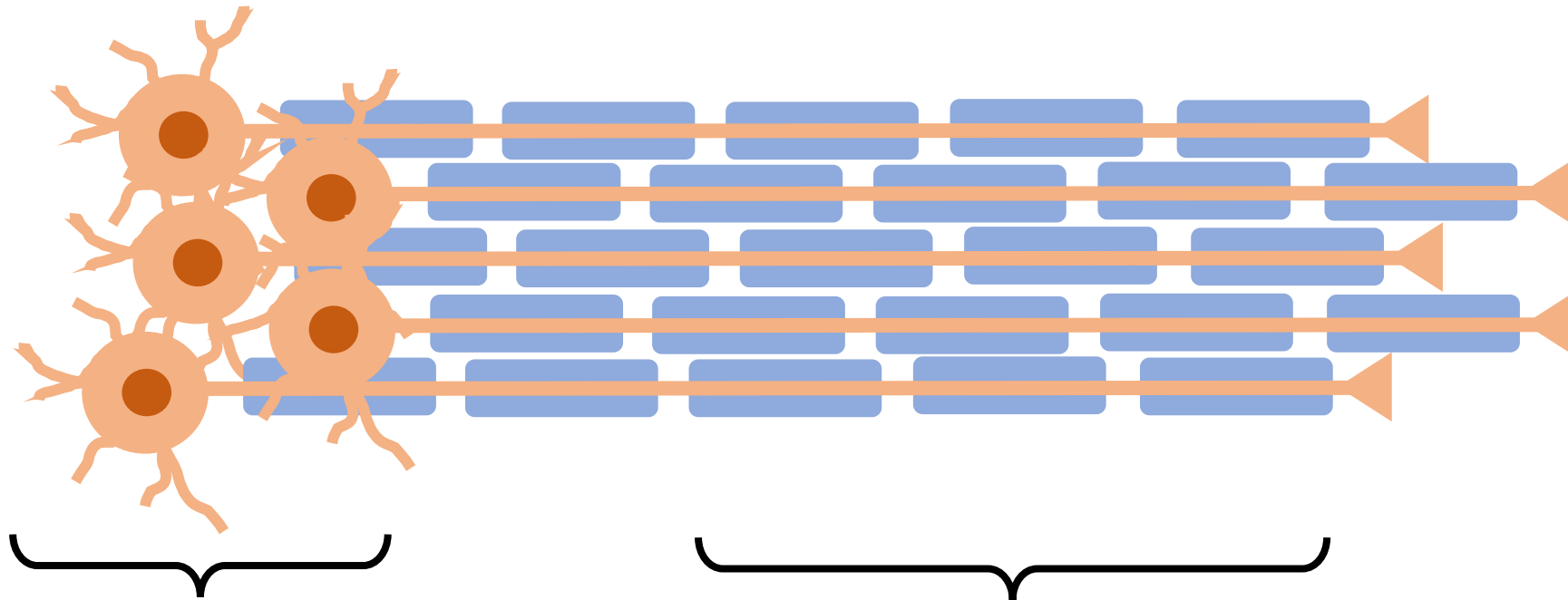
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Building Block of the Nervous System: The Neuron



- Neurons, also called nerve cells, are fundamental building blocks of the nervous system. The diameter of a neuronal soma or cell body is between 10 and 100 micrometers. Axons can cover distances from a few micrometers in the immediate vicinity to slightly more than 1 meter (more than 3 feet). Most neuronal axons are myelinated.

Gray and White Matter in the CNS

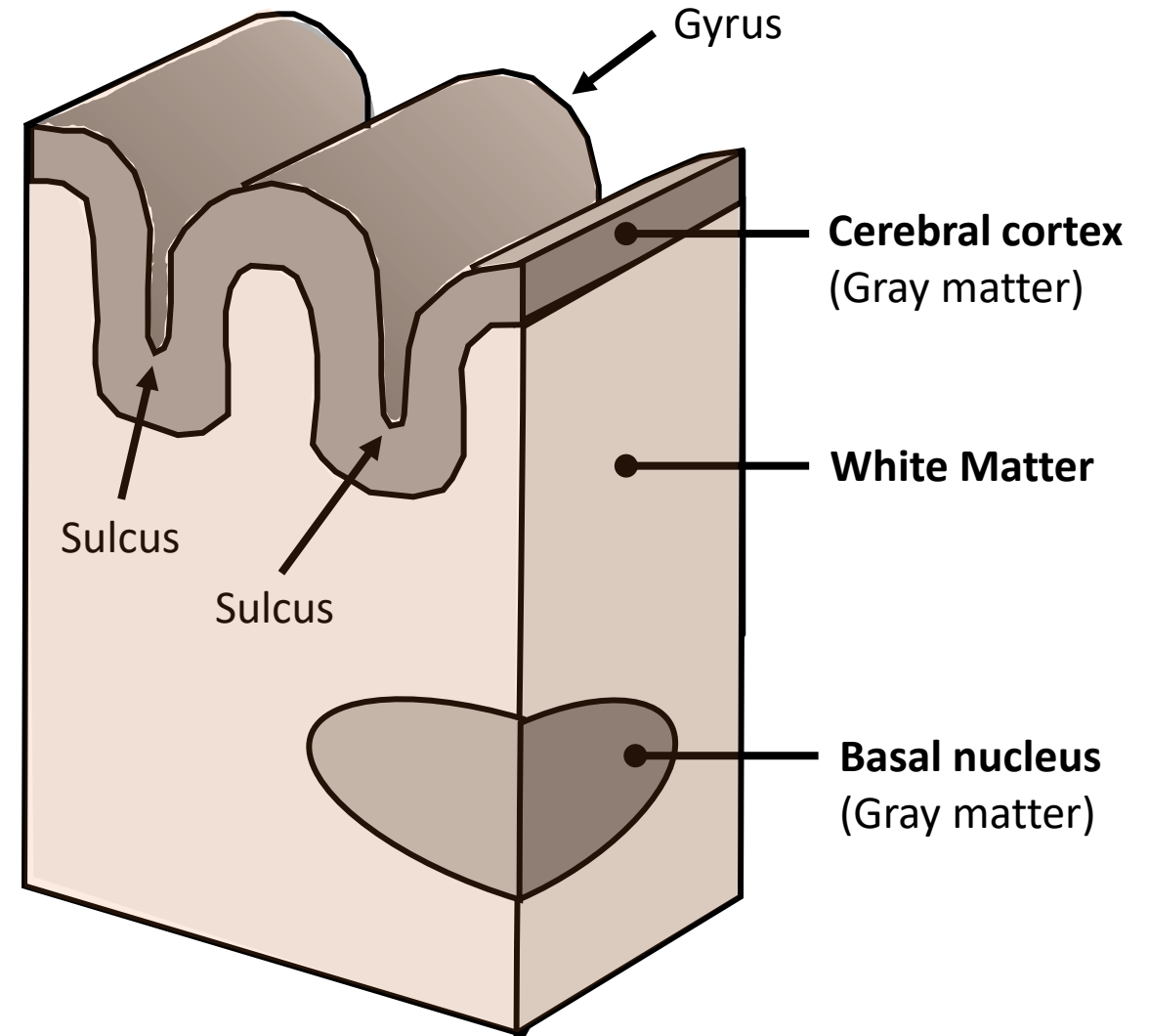


- **Gray matter** gets its darker tone from a high concentration of neuronal cell bodies, based on their high RNA content, and a high density of capillaries supplying them with oxygen and nutrients.

- **White matter** gets its paler tone from a high amount of myelin, a fatty substance, which provides electrical insulation of axons, improving their conduction velocity.

Cerebral Hemispheres – Gray and White Matter

- In the cerebral hemispheres, most of the gray matter is located close to the surface, in outermost layer called the cerebral cortex
- Gyri (elevations) and sulci (grooves) are the two main types of surface folds of the cerebral cortex
- Some of the gray matter forms distinct nuclei in the deeper layers of the white matter
- The deeper nuclei are called basal ganglia or basal nuclei



Gray Matter Terminology

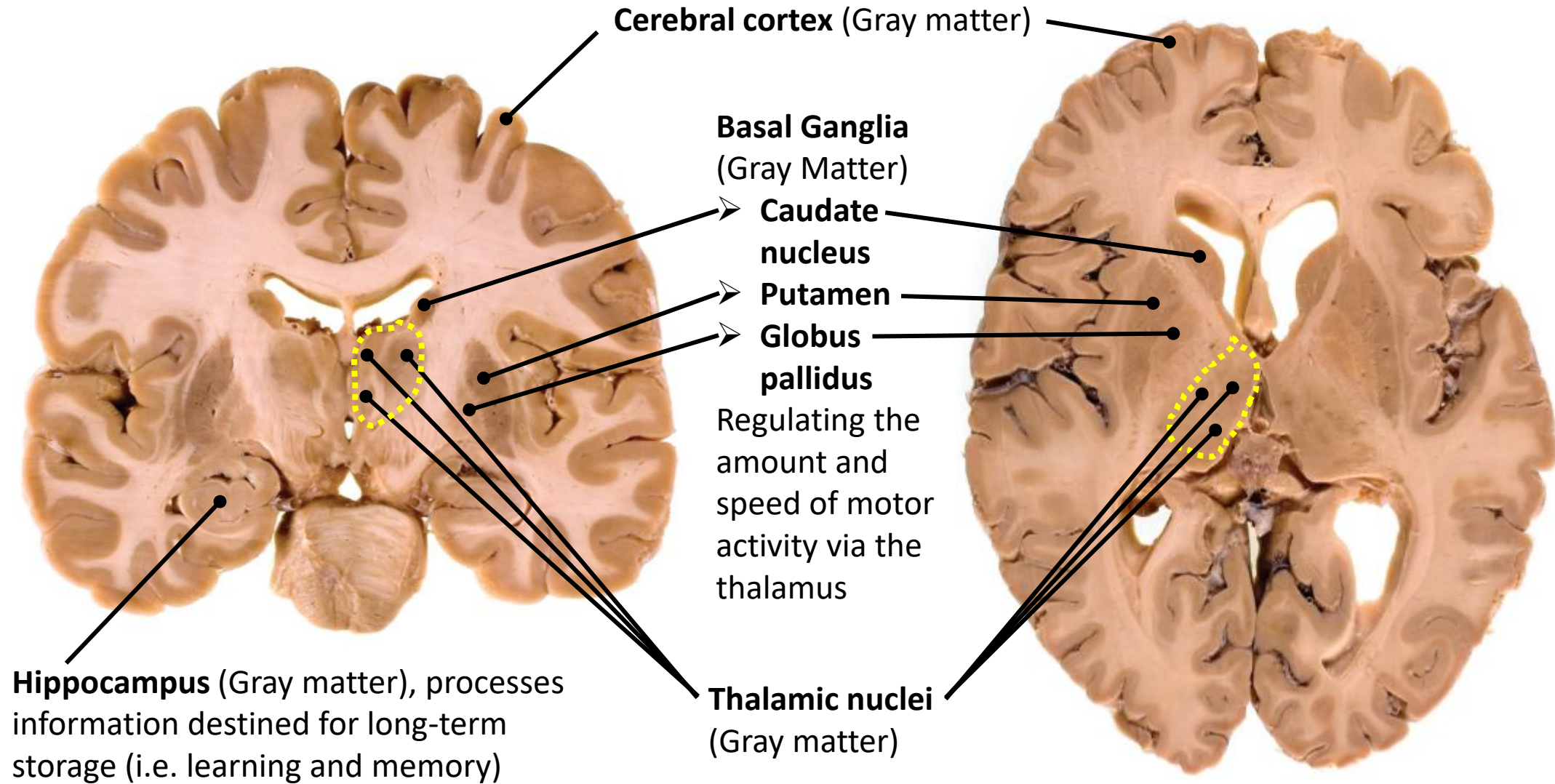
- **Central Nervous System (CNS)**

- **Cerebral cortex:** superficial component of the gray matter of the cerebral hemisphere
- **Nucleus:** most common term for distinct lump of gray matter, usually with a distinct function (examples: red nucleus, substantia nigra)
- **Reticular formation:** describes network of smaller nuclei of the brainstem connected by fiber tracts, which gives it the appearance of a fishing net (example: pontine paramedian reticular formation)

- **Peripheral Nervous System (PNS)**

- **Ganglion:** cluster of neurons usually surrounded by non-neuronal tissue; can be either somatic or visceral (examples: dorsal root ganglia (somatic), sympathetic chain ganglia (visceral))

Gray Matter in Coronal and Horizontal Sections



White Matter: Categories of Fiber Tracts

- **Projection Fibers**

- Connect higher (such as the cerebral cortex) and lower (such as the spinal cord) elements of the CNS along the rostro-caudal axis
- The direction of the information flow can be in both directions, either **ascending** or **descending**
- Examples are the **internal capsule** and the **corona radiata**

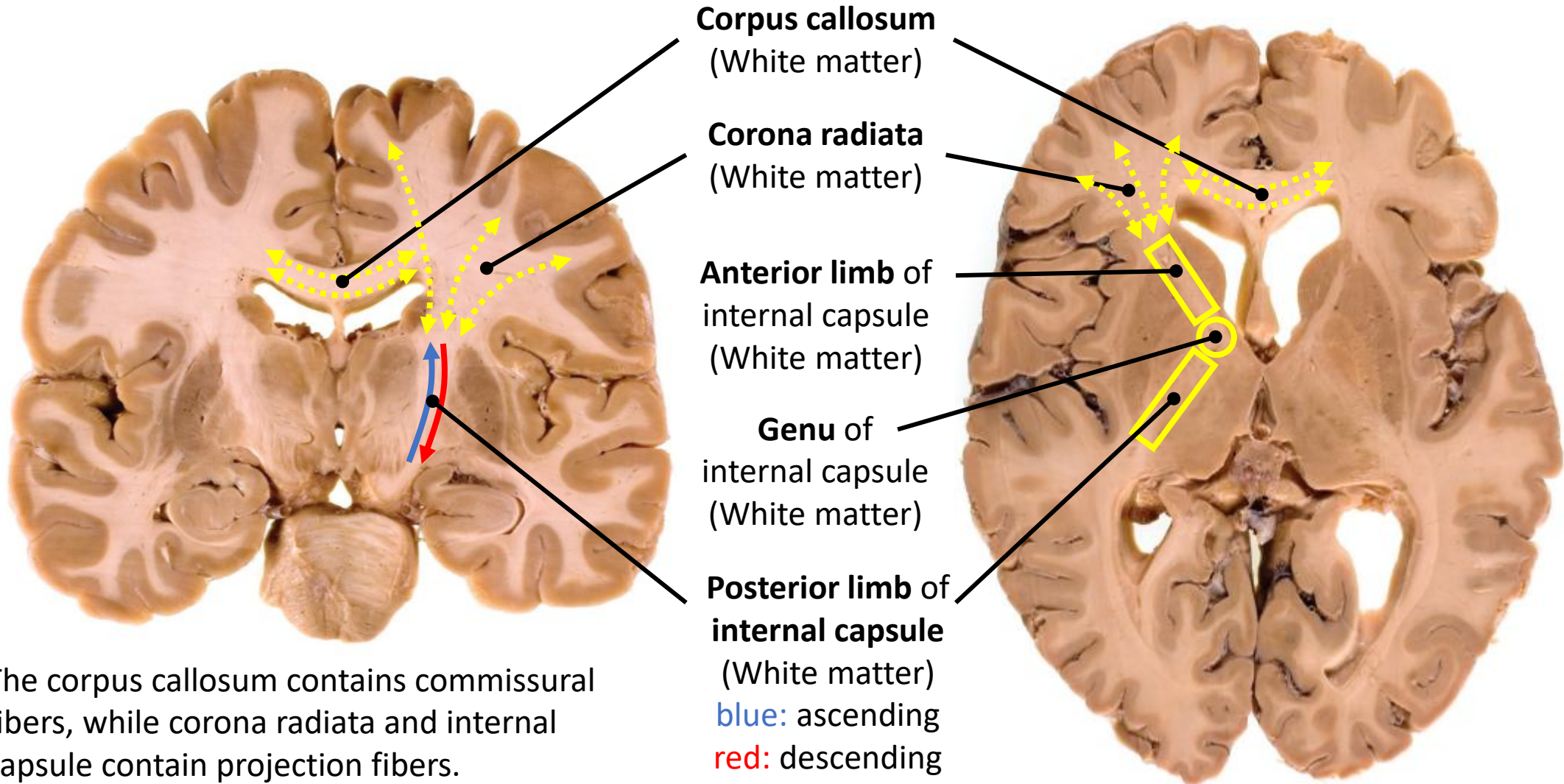
- **Commissural Fibers**

- Connect the two hemispheres of the cerebral cortex to each other
- The **corpus callosum** is the largest commissural fiber tract

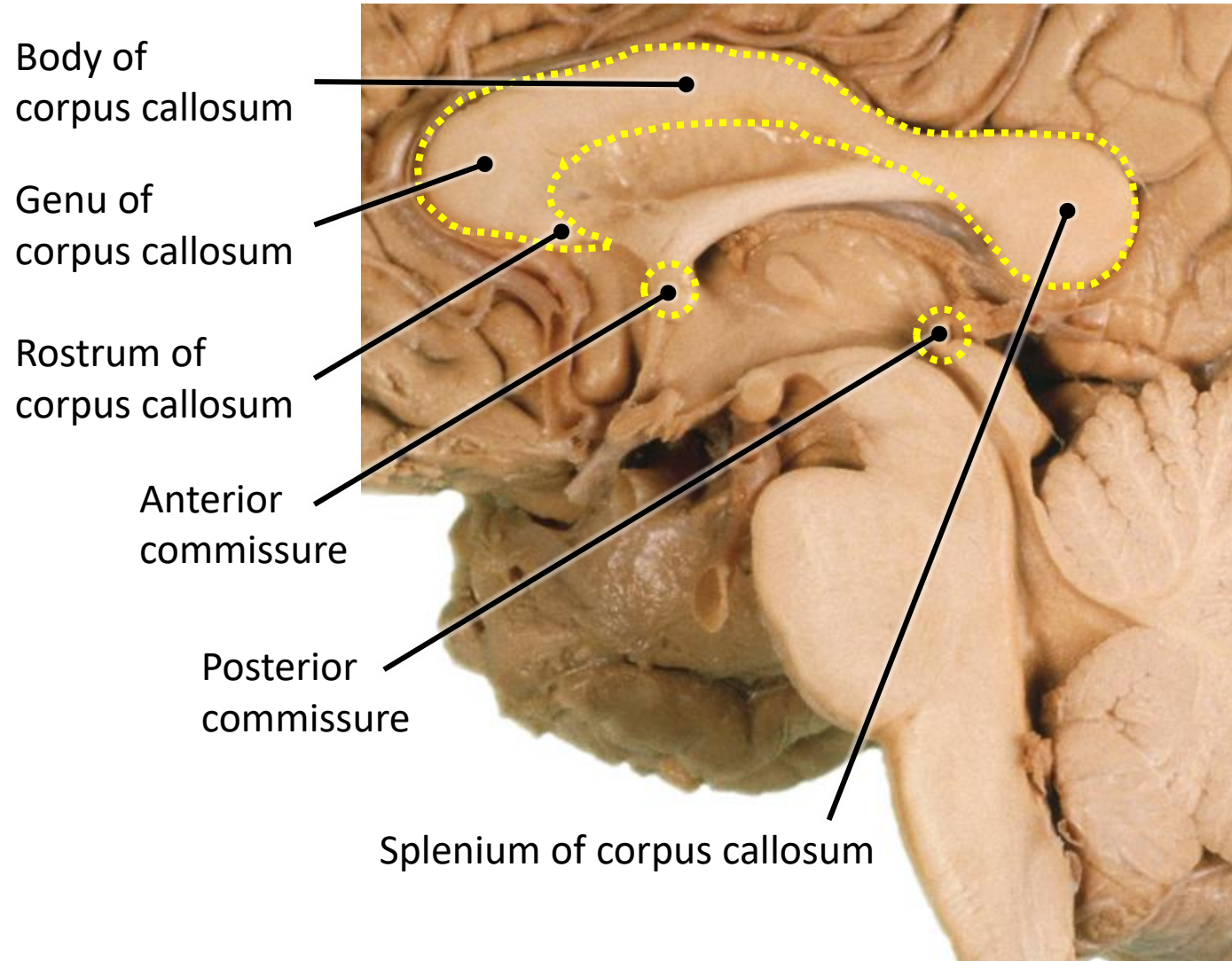
- **Association Fibers**

- Connect different parts of the cerebral cortex of one hemisphere
- An example is the **arcuate fasciculus**, which connects Broca's area (language production) and Wernicke's area (understanding of language) in the same hemisphere

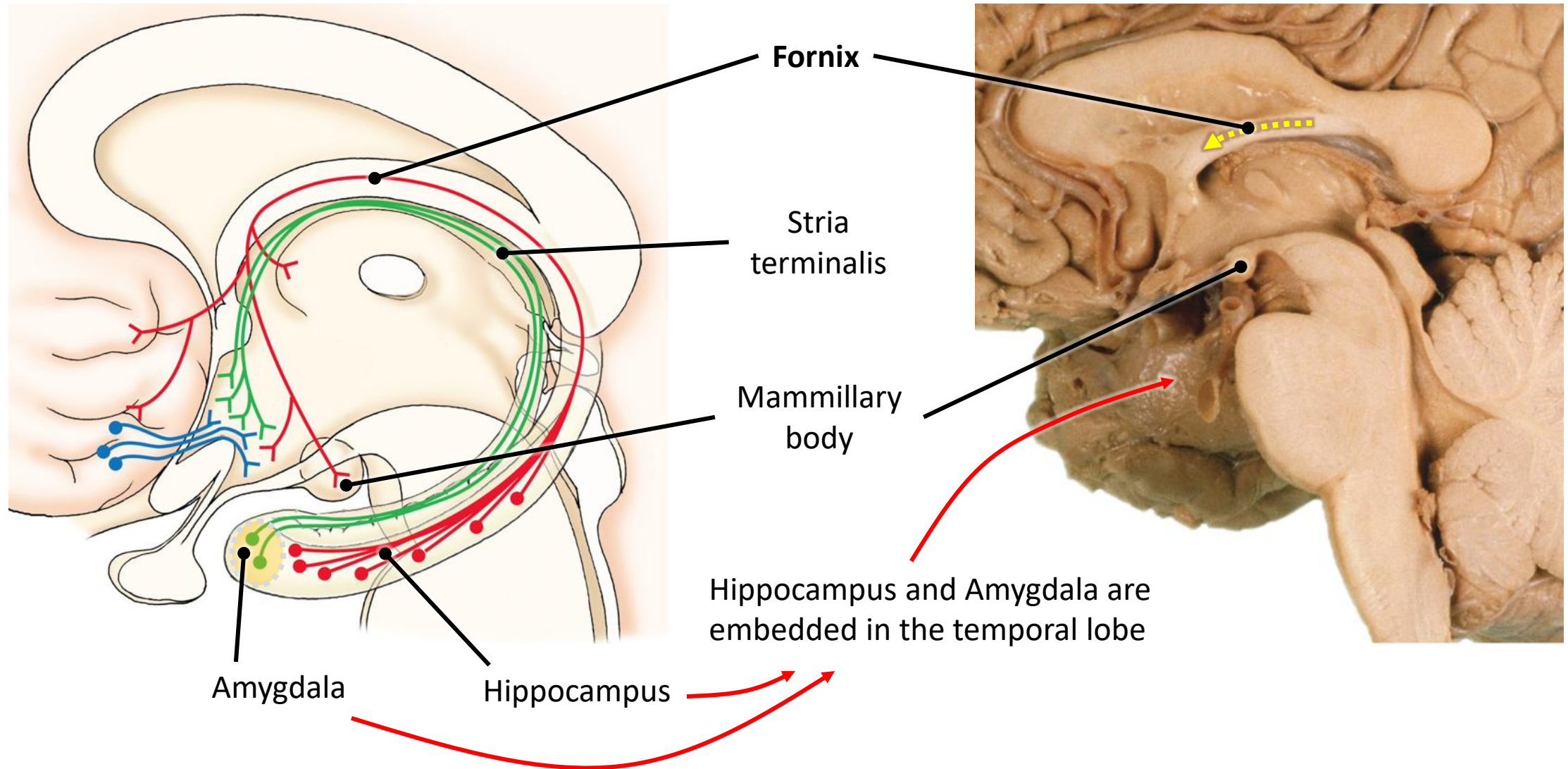
White Matter in Coronal and Horizontal Sections



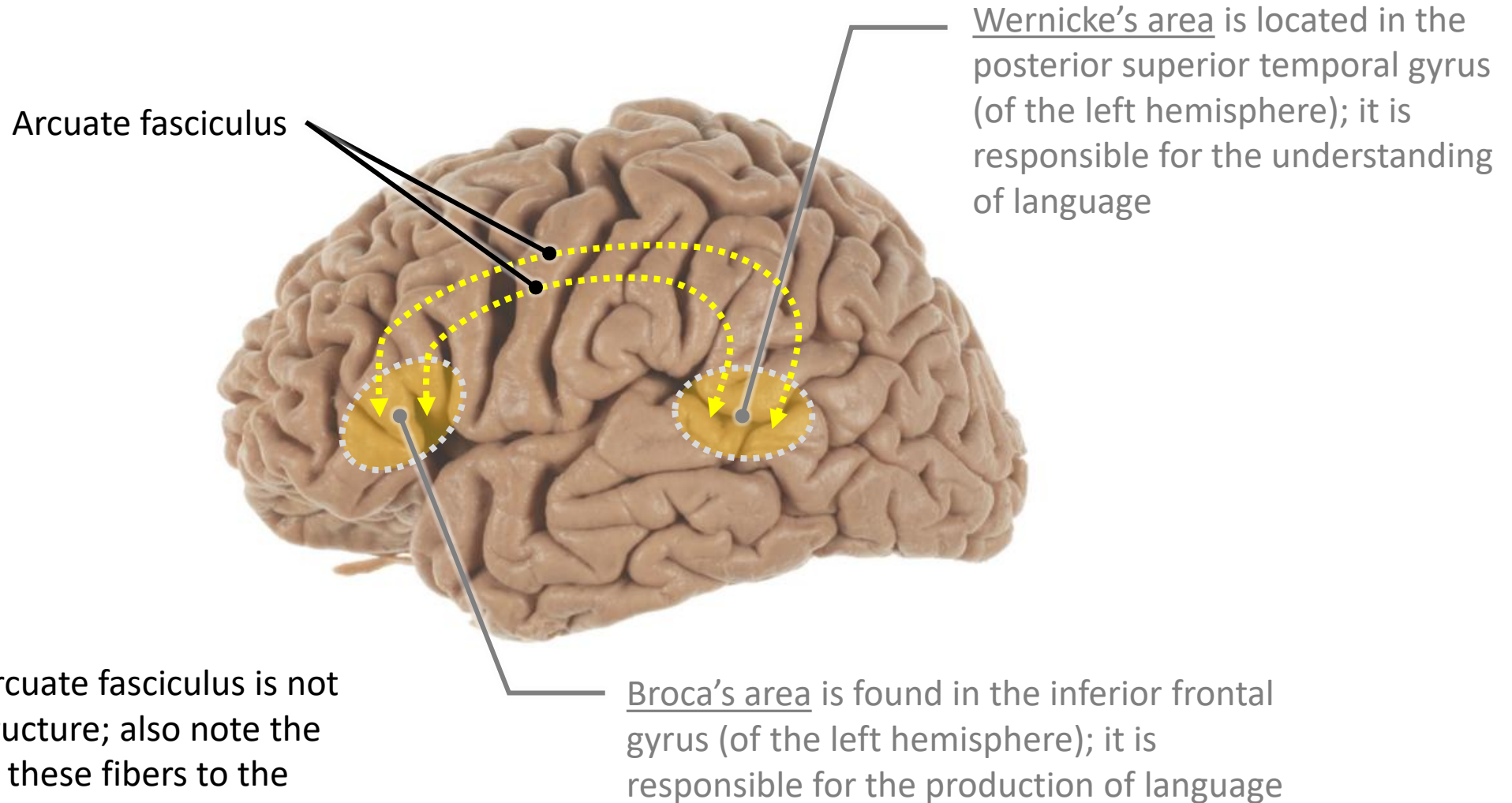
Commissures connect Left and Right Hemispheres



Projection Fibers of the Fornix connect the Hippocampus with other Brain Regions



Association Fibers of the Arcuate Fasciculus connect Broca's and Wernicke's Areas



Note: The arcuate fasciculus is not a surface structure; also note the proximity of these fibers to the supramarginal and angular gyri

White Matter Terminology

- **Central Nervous System (CNS)**

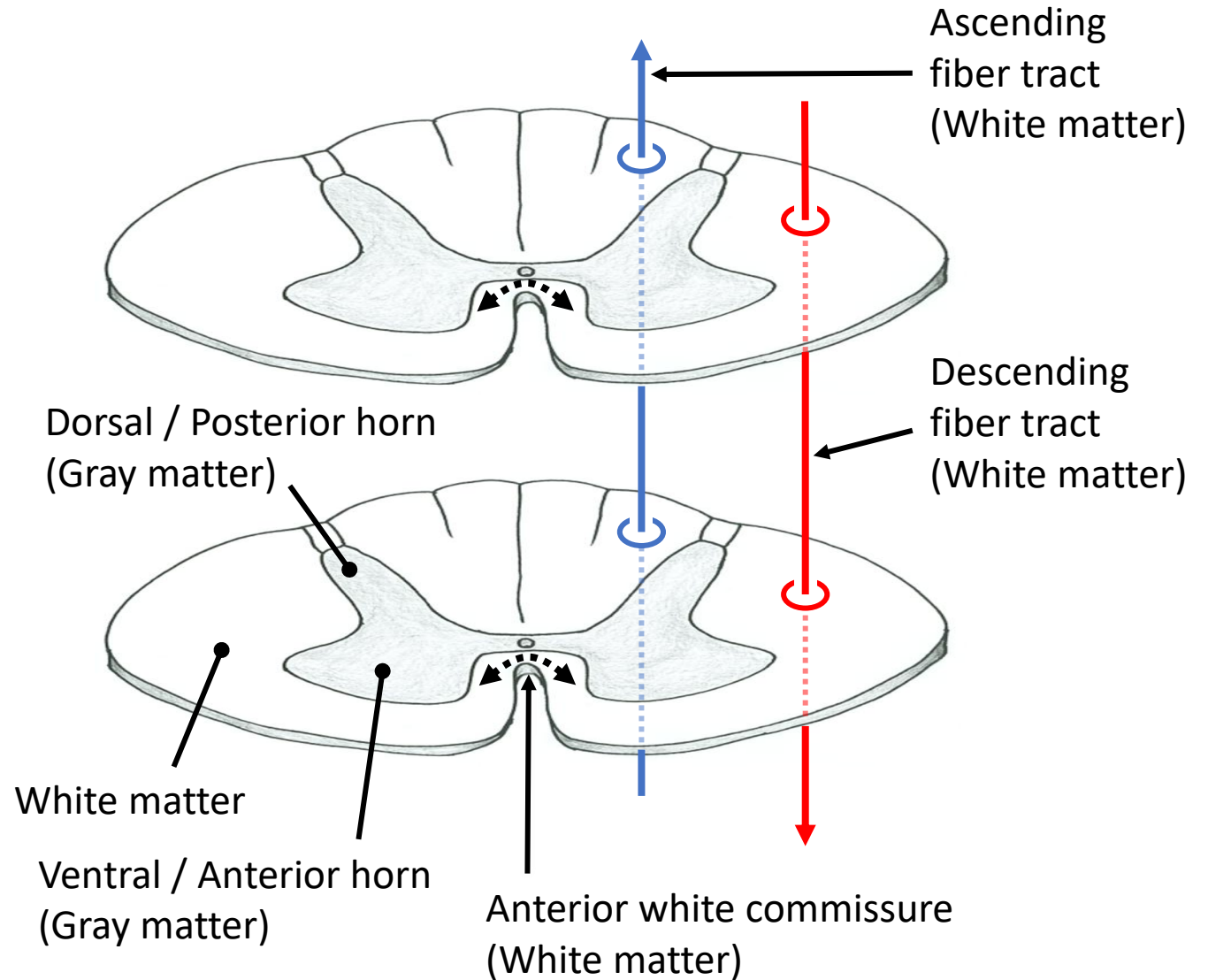
- In addition to the terms already introduced above, there are other common terms for white matter tracts in the CNS
- **Fasciculus:** bundle of nerve fibers
- **Fornix:** arched, C-shaped bundle of nerve fibers
- **Funiculus:** bundle of nerve fibers
- **Lemniscus:** a thin, ribbon-like band of nerve fibers
- **Peduncle:** a stalk-like bundle of nerve fibers, connecting one part of the brain with another part of the brain
- **Stria:** a narrow, streak-like bundle of nerve fibers

- **Peripheral Nervous System (PNS)**

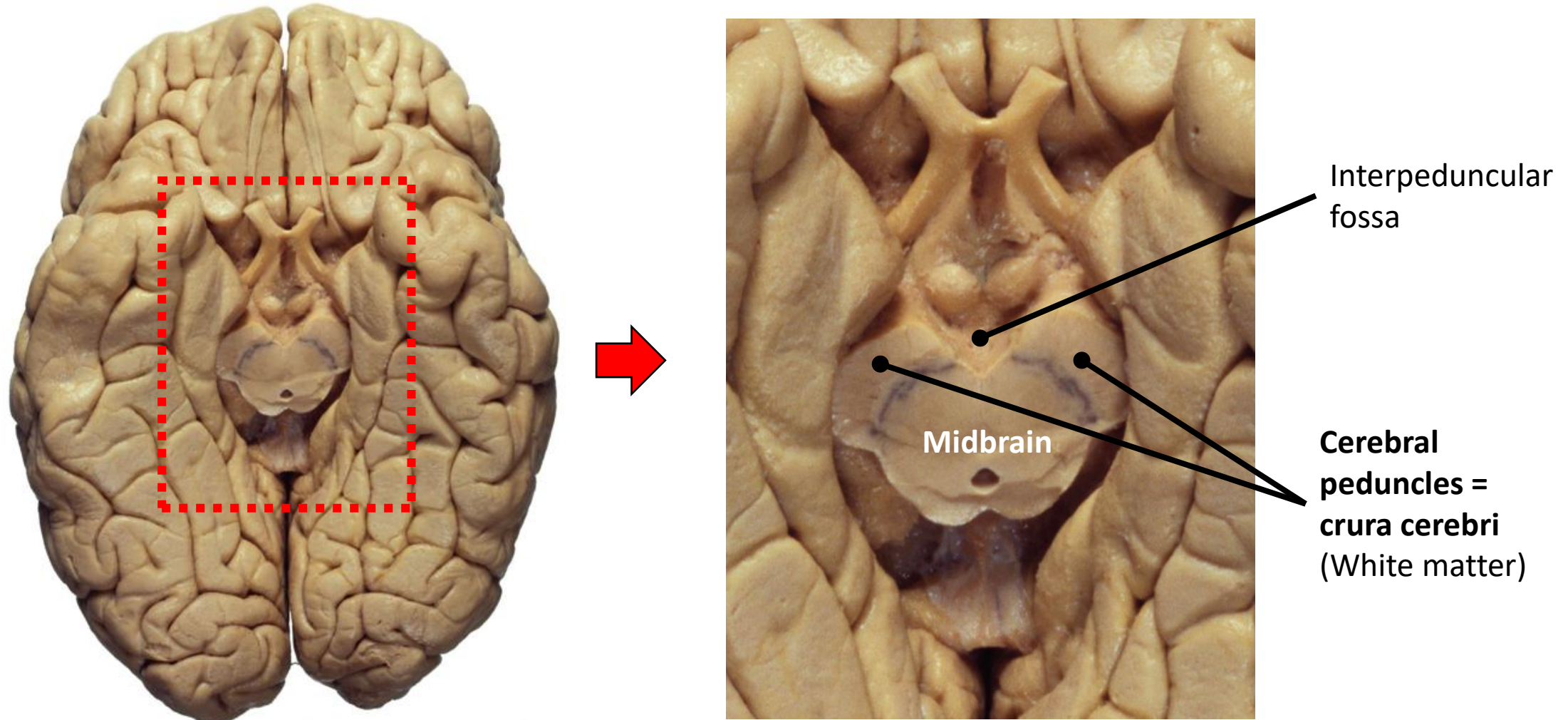
- **Nerve:** bundle of axons covered by connective tissue in the peripheral nervous system (examples: cranial nerves (such as the optic nerve), spinal nerves (such as L2), more distal branches (such as the ulnar nerve))

Gray and White Matter in Axial Spinal Cord Sections

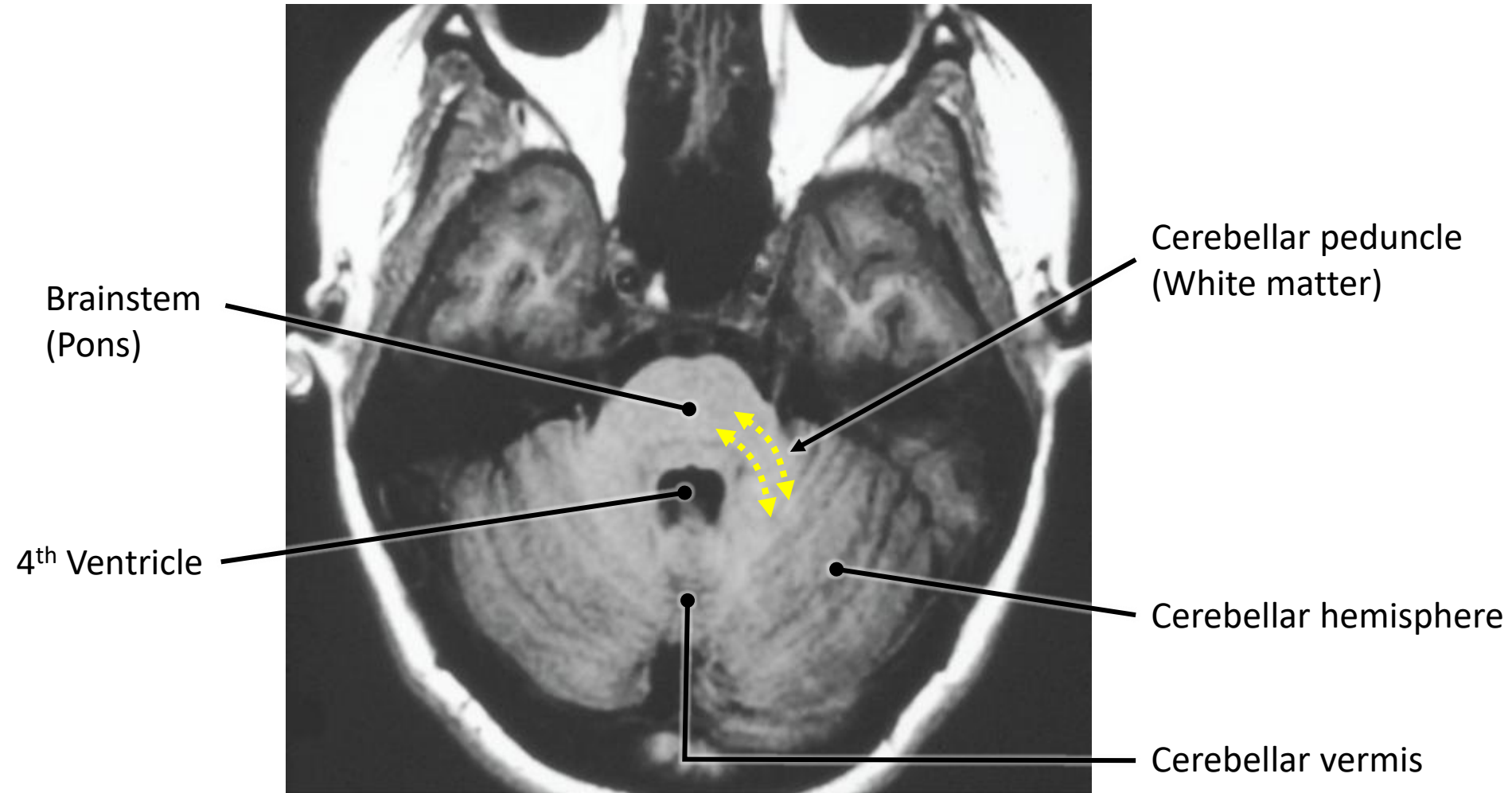
- **Gray matter** is in the center of the cord, surrounded by white matter
- **Dorsal horns** receive somatosensory input, and **ventral horns** produce motor output
- Lateral horns of gray matter are found in the thoracic and lumbar spinal cord, as part of autonomics
- **White matter** contains **ascending and descending fiber tracts**



Cerebral Peduncles are Part of the Midbrain and connect the Cerebral Hemispheres with the Rest of the Brainstem



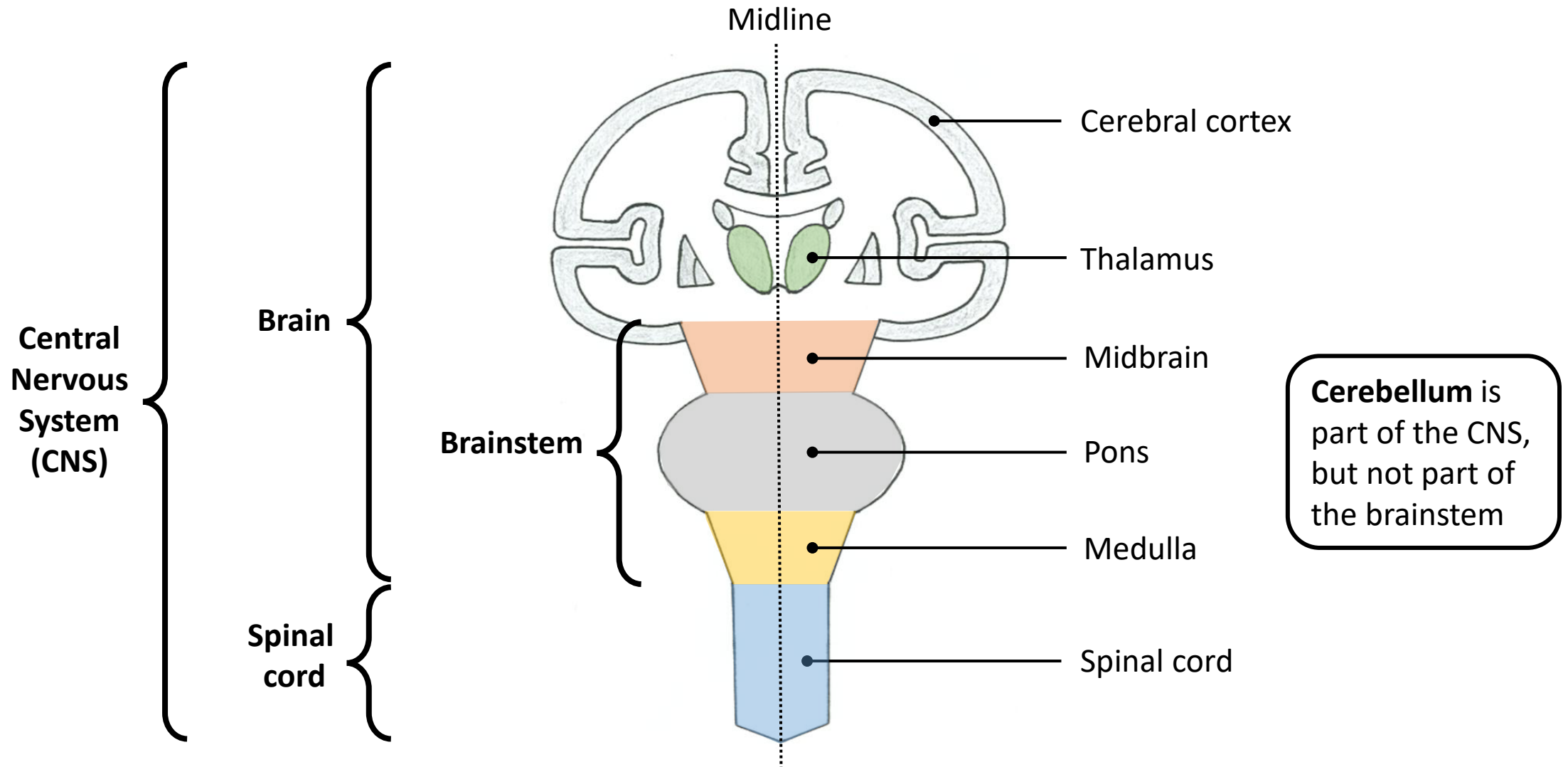
Cerebellar Peduncles connect Brainstem and Cerebellum



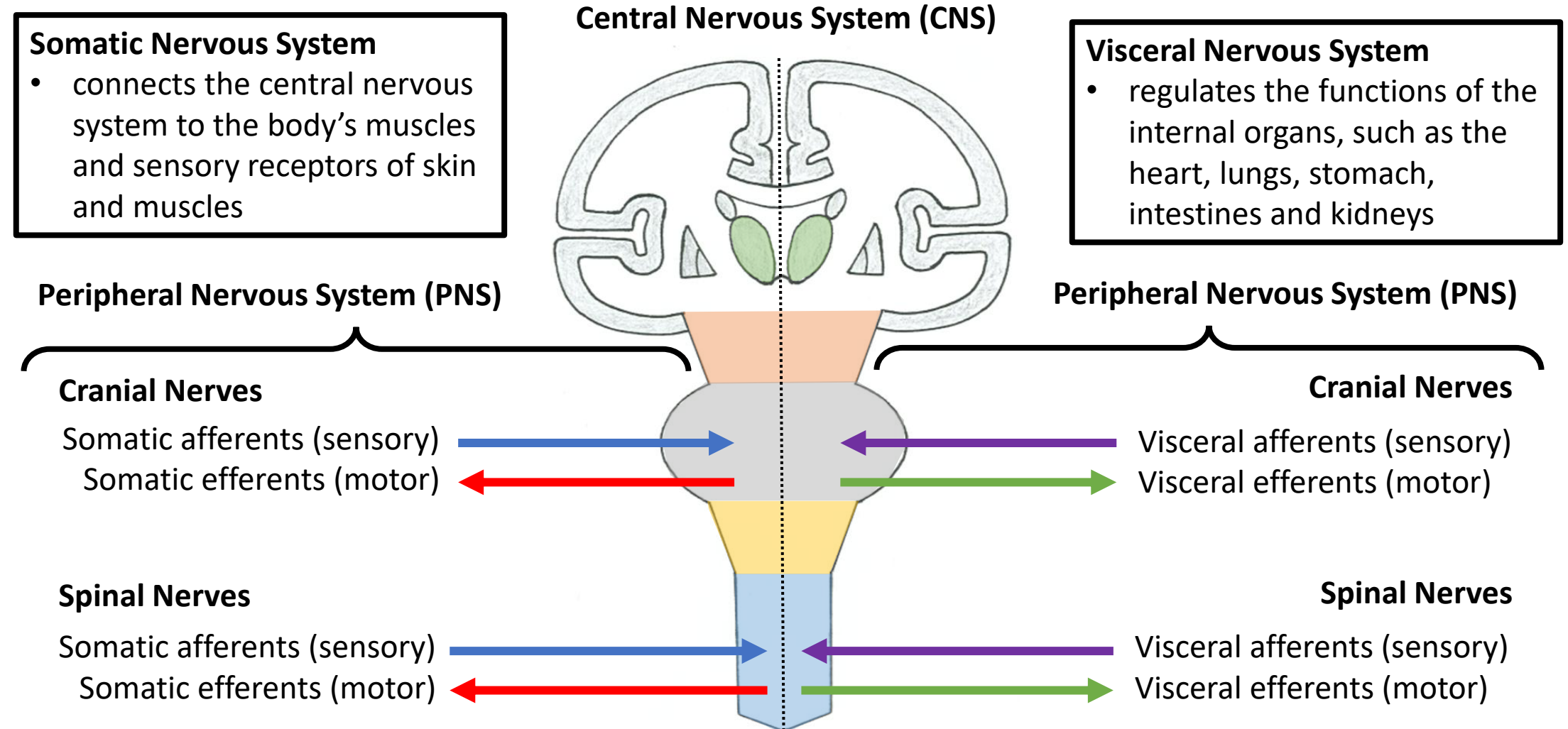
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Central Nervous System (CNS) Schematic



Communications between CNS and PNS – Afferents and Efferents



Note: Cranial and spinal nerves appear in pairs, one on each side, containing somatic or visceral fibers, or both.

Afferents and Efferents – Terminology Overview

Somatic Nervous System

- Somatic Afferents (towards CNS)
 - Somatosensory System
 - Examples: touch, vibration, proprioception, pain and temperature of the face (via cranial nerves) and the body (via spinal nerves)
 - Special Senses
 - Examples: vision and hearing (via cranial nerves)
- Somatic Efferents (away from CNS)
 - Somatic Motor System
 - Examples: eye movements and tongue movements (via cranial nerves); limb and body movements (via spinal nerves)

Visceral Nervous System

- Visceral Afferents (towards CNS)
 - General Visceral Afferents
 - Examples: sensory information from the internal organs (may follow sympathetic or parasympathetic pathways)
 - Special Visceral Afferents
 - Examples: smell and taste (via cranial nerves)
- Visceral Efferents (away from CNS)
 - Sympathetic Nervous System
 - Thoraco-lumbar spinal cord origin; example: dilation of the pupil
 - Parasympathetic Nervous System
 - Examples: constriction of the pupil (via cranial nerves); emptying of bladder (via spinal nerves)

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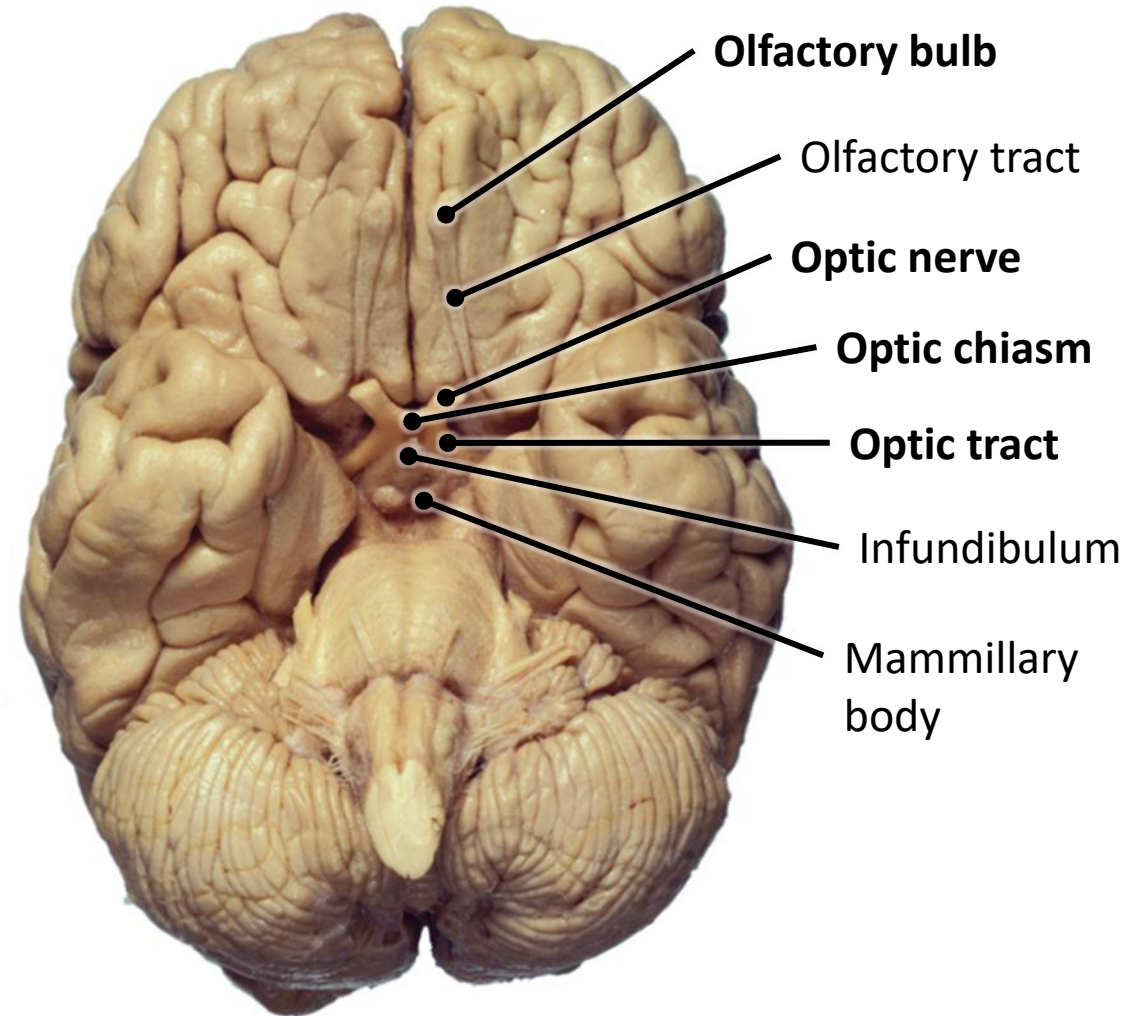
Cranial Nerves of the Telencephalon and Diencephalon

- **CN I: Olfactory nerve**

- Location:
 - Running through cribriform plate, connecting olfactory epithelium with olfactory bulb/tract/telencephalon
- Function(s):
 - Visceral afferents (special senses): olfaction (smell)

- **CN II: Optic nerve**

- Location:
 - Connecting the retina of the eye with the diencephalon
- Function(s):
 - Somatic afferents (special senses): vision

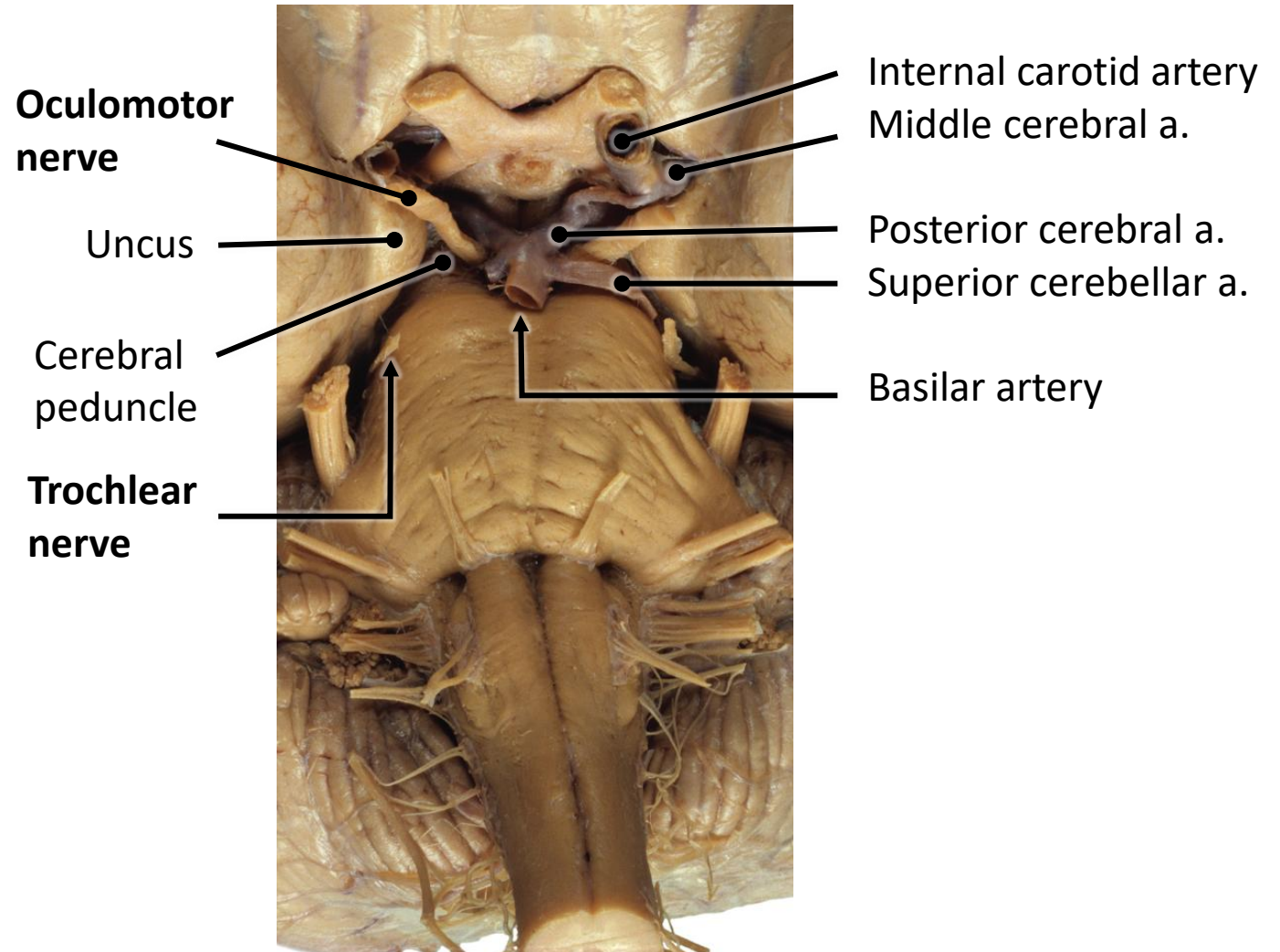


Cranial Nerves of the Midbrain (Anterior View)

- **CN III: Oculomotor nerve**

- Location:
 - Medial upper midbrain, level of the superior colliculi
- Function(s):
 - Somatic efferents: eye movements
 - Visceral efferents: near accommodation and pupillary constriction

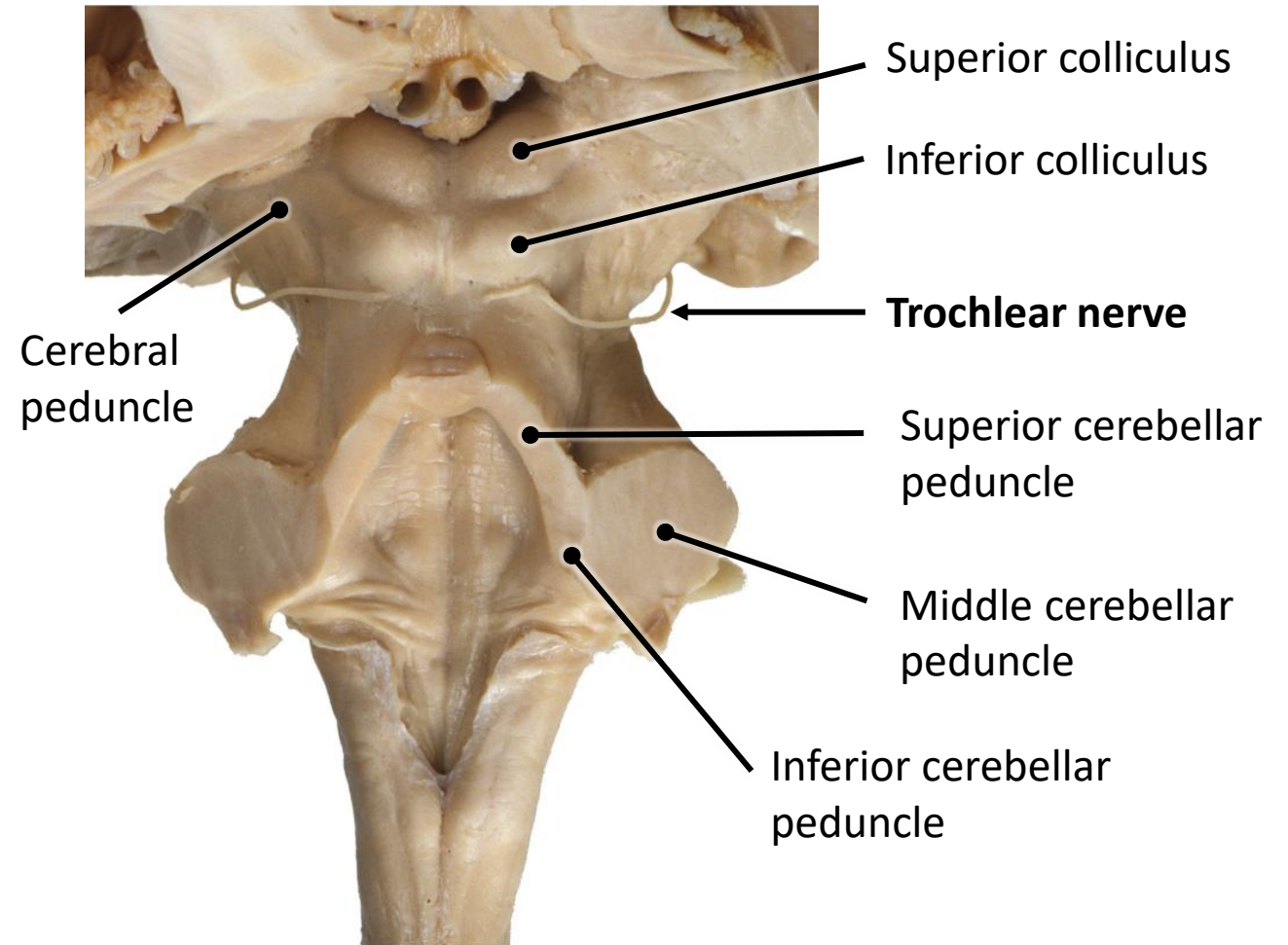
Please note the proximity of the uncus to the midbrain structures (will become relevant in clinical scenarios later)



Cranial Nerves of the Midbrain (Posterior View)

- **CN IV: Trochlear nerve**

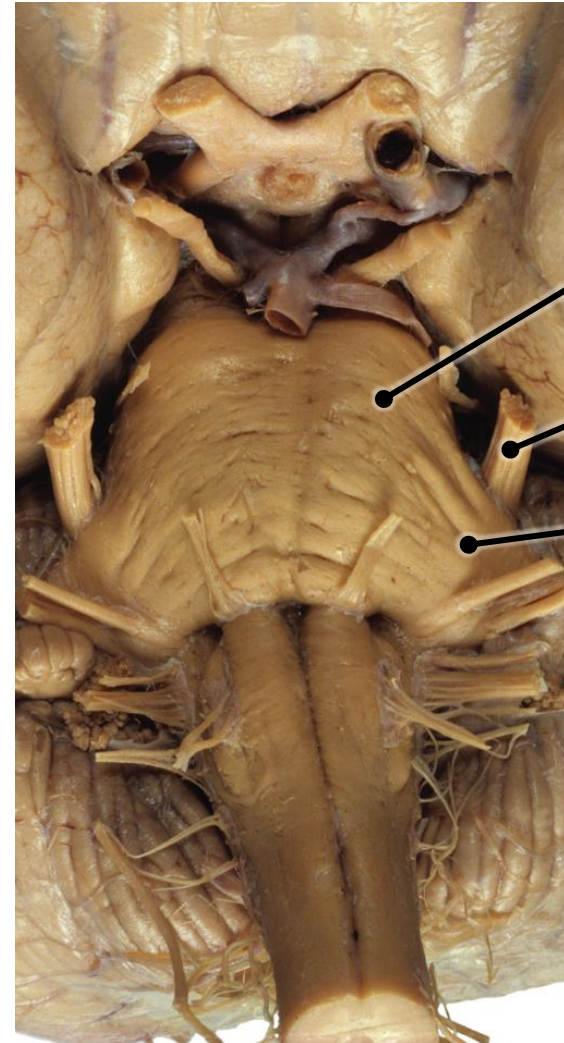
- Location:
 - Only cranial nerve at the posterior brainstem
 - Medial lower midbrain, close to inferior colliculi
- Function(s):
 - Somatic efferents: eye movements



Cranial Nerves of the Pons (Anterior View)

- **CN V: Trigeminal nerve**

- Location:
 - Lateral aspect of mid-pons
- Function(s):
 - Somatic afferents: somatosensations (touch, vibration, proprioception, pain and temperature) of the face
 - Somatic efferents: motor innervation of muscles of mastication



Basilar pons

Trigeminal nerve

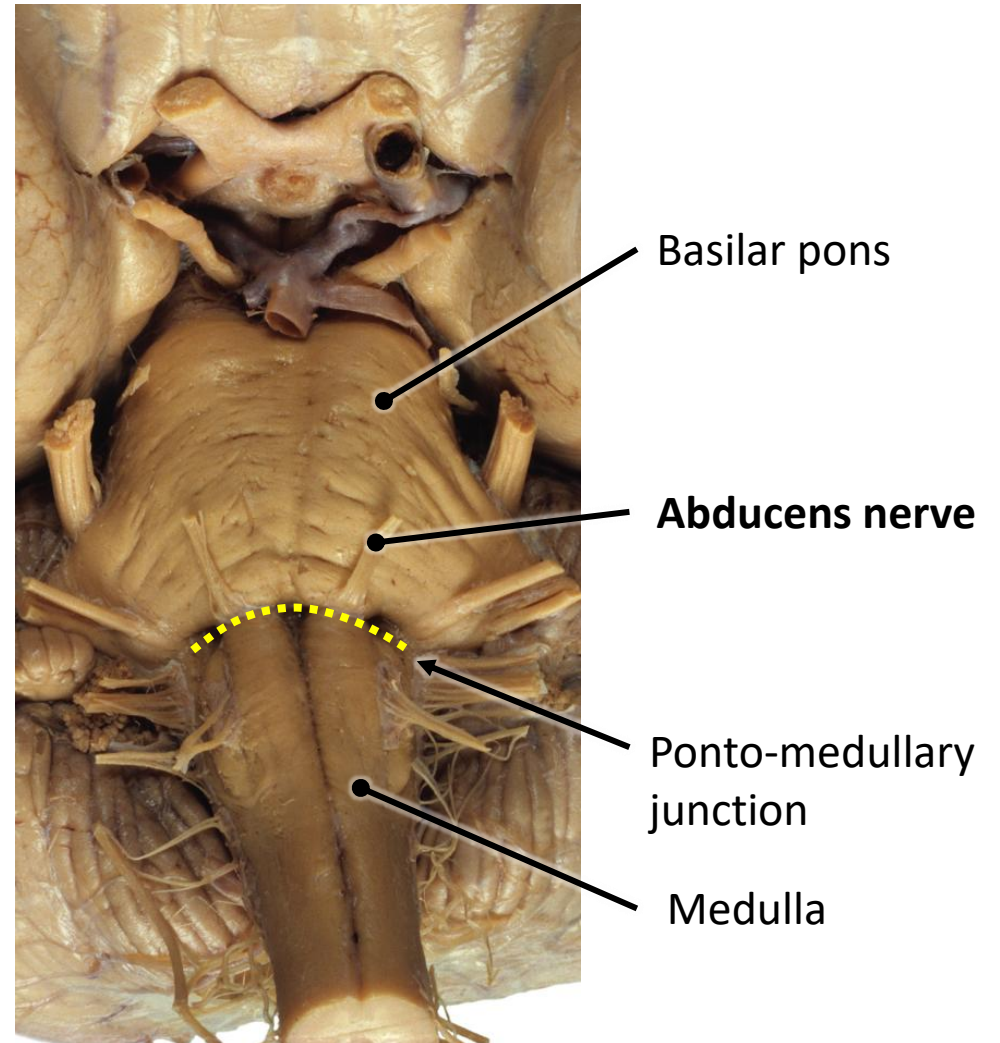
Middle cerebellar peduncle

Cranial Nerves of the Ponto-Medullary Junction

- **CN VI: Abducens nerve**

- Location:
 - Medial ponto-medullary junction
- Function(s):
 - Somatic efferents: eye movements

Note: As a general principle, based on the unfolding of basal plate and alar plate during development of the brainstem, the location **motor** nuclei is more **medial**, while the location of **sensory** nuclei is more **sideways**.

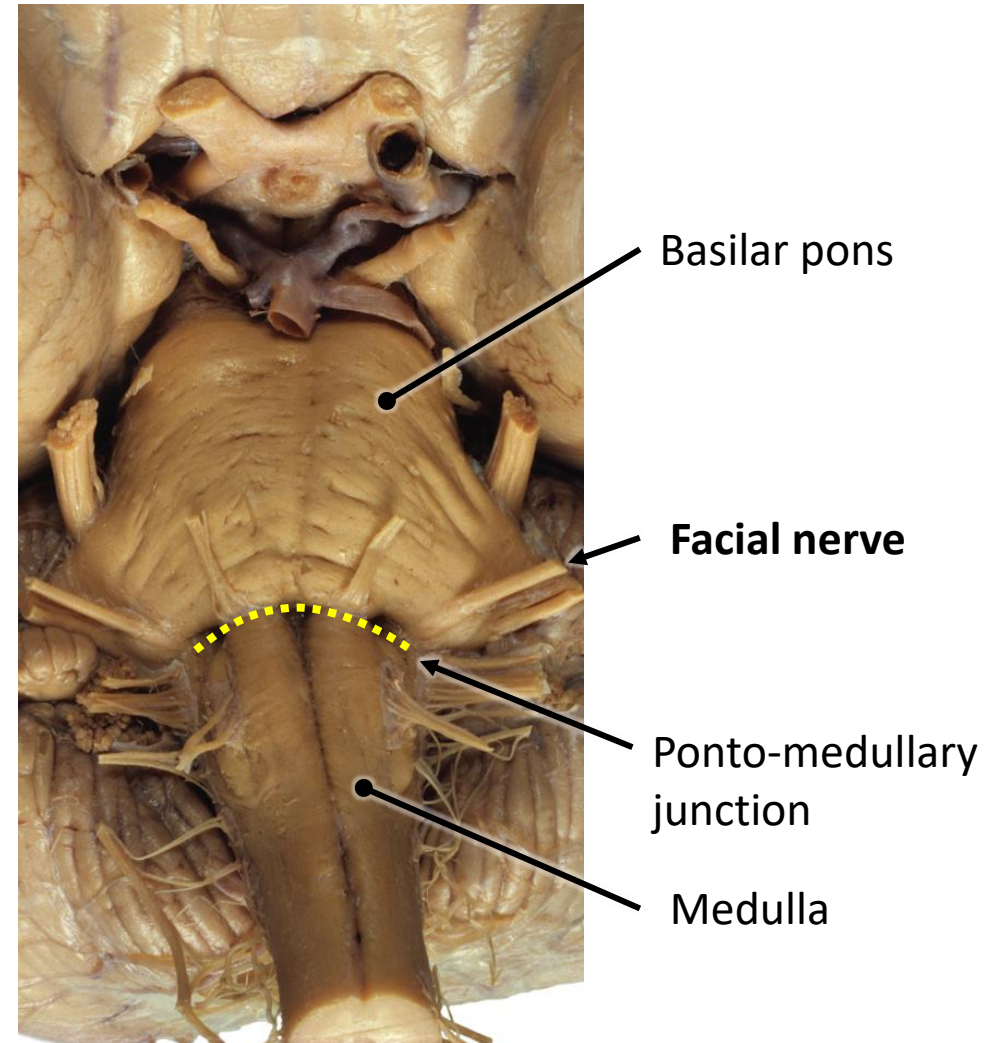


Cranial Nerves of the Ponto-Medullary Junction

- **CN VII: Facial nerve**

- Location:
 - Intermediate ponto-medullary junction
- Function(s):
 - Somatic efferents: motor innervation of muscles of facial expression
 - Visceral afferents (special senses): gustation (taste) of the anterior tongue
 - ... *)

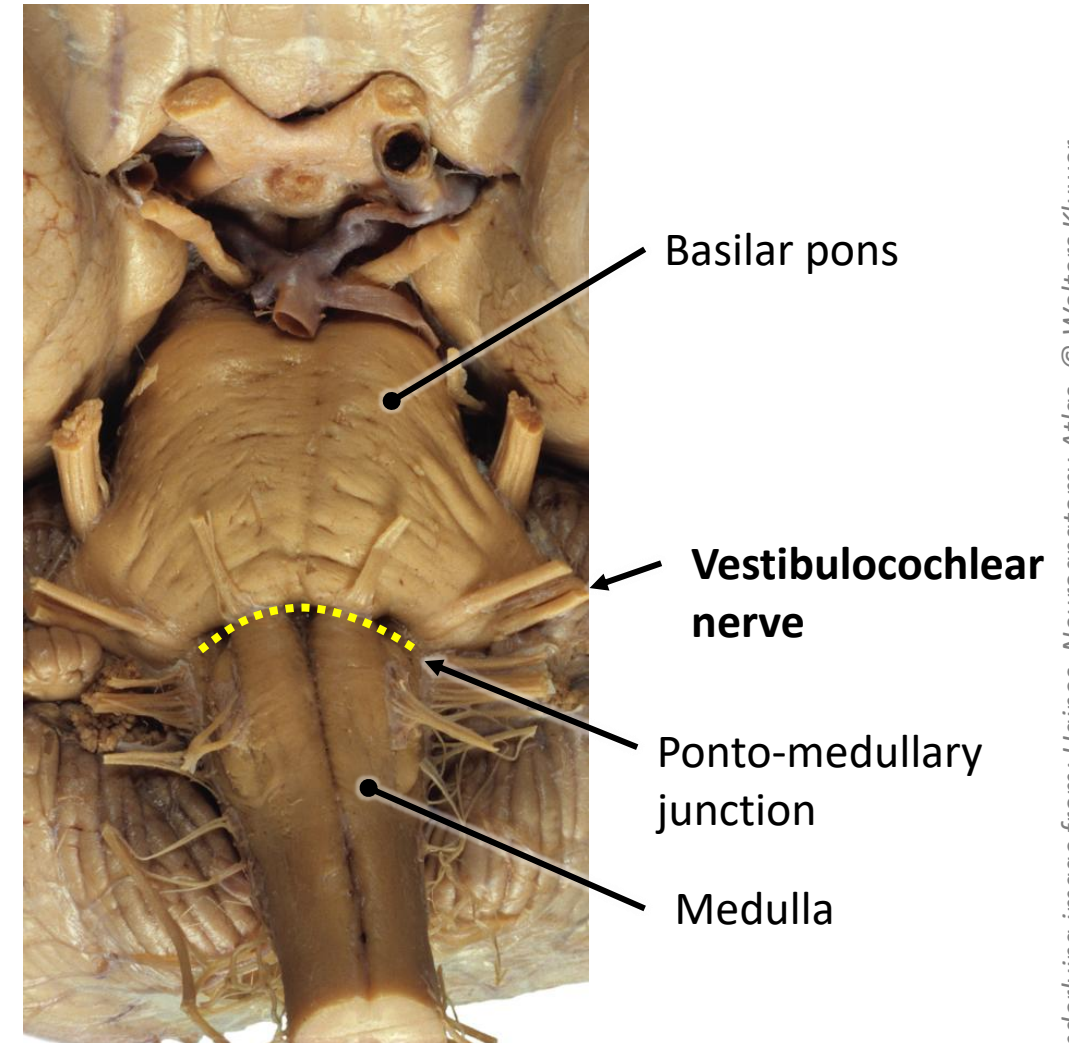
*) Note: In this lecture, we focus on a few key components and functions of cranial nerves; more details will be added in due course.



Cranial Nerves of the Ponto-Medullary Junction

- **CN VIII: Vestibulocochlear nerve**

- Location:
 - Lateral ponto-medullary junction
- Function(s):
 - Somatic afferents (cochlear portion): audition (hearing)
 - Somatic afferents (vestibular portion): balance and equilibrium

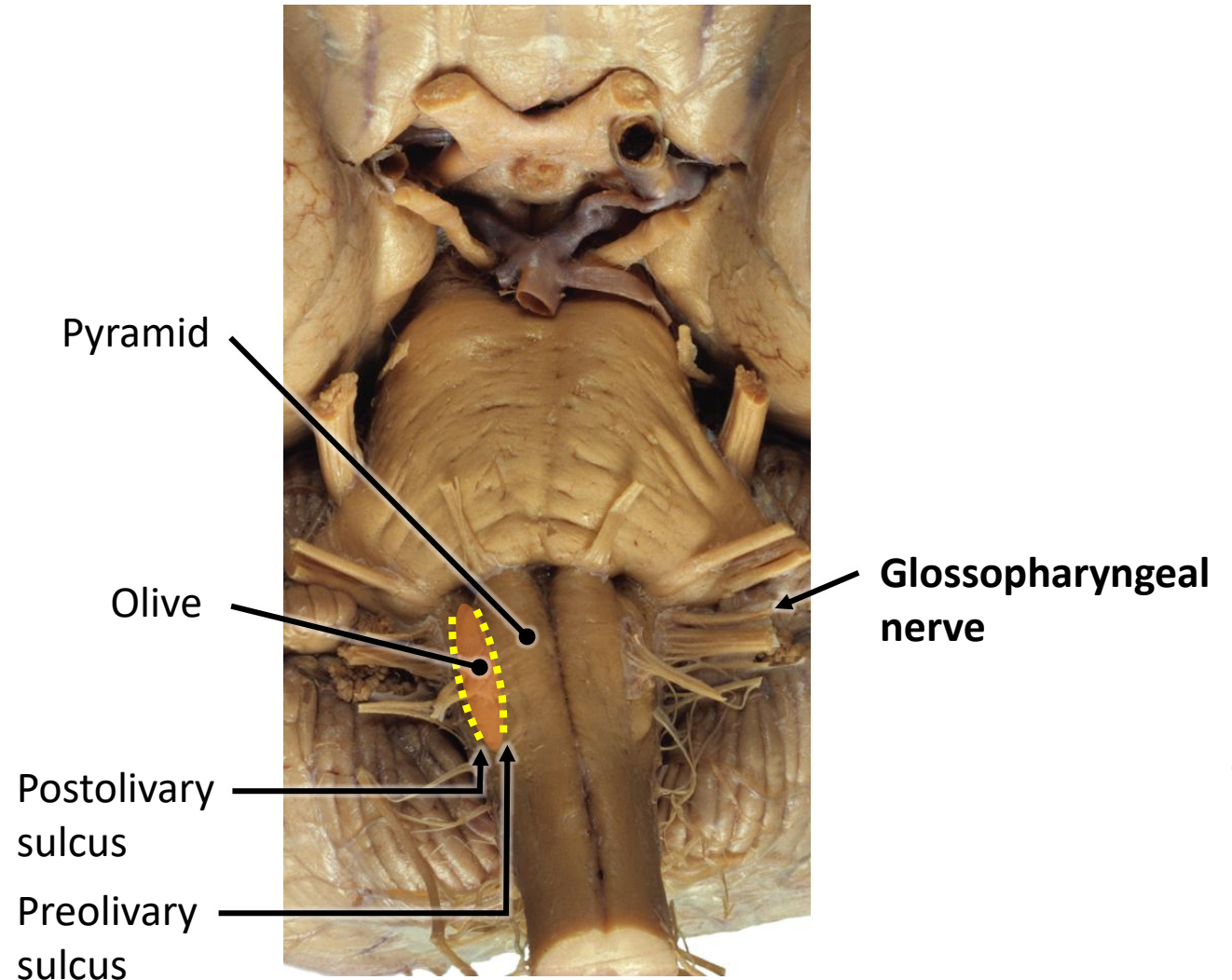


Underlying image from: Haines, Neuroanatomy Atlas, © Wolters Kluwer
<https://sgu.idm.oclc.org/login?url=https://meded.lwwhealthlibrary.com/content.aspx?bookid=2473>, Figure 2-20

Cranial Nerves of the Medulla (Anterior View)

- **CN IX: Glossopharyngeal nerve**

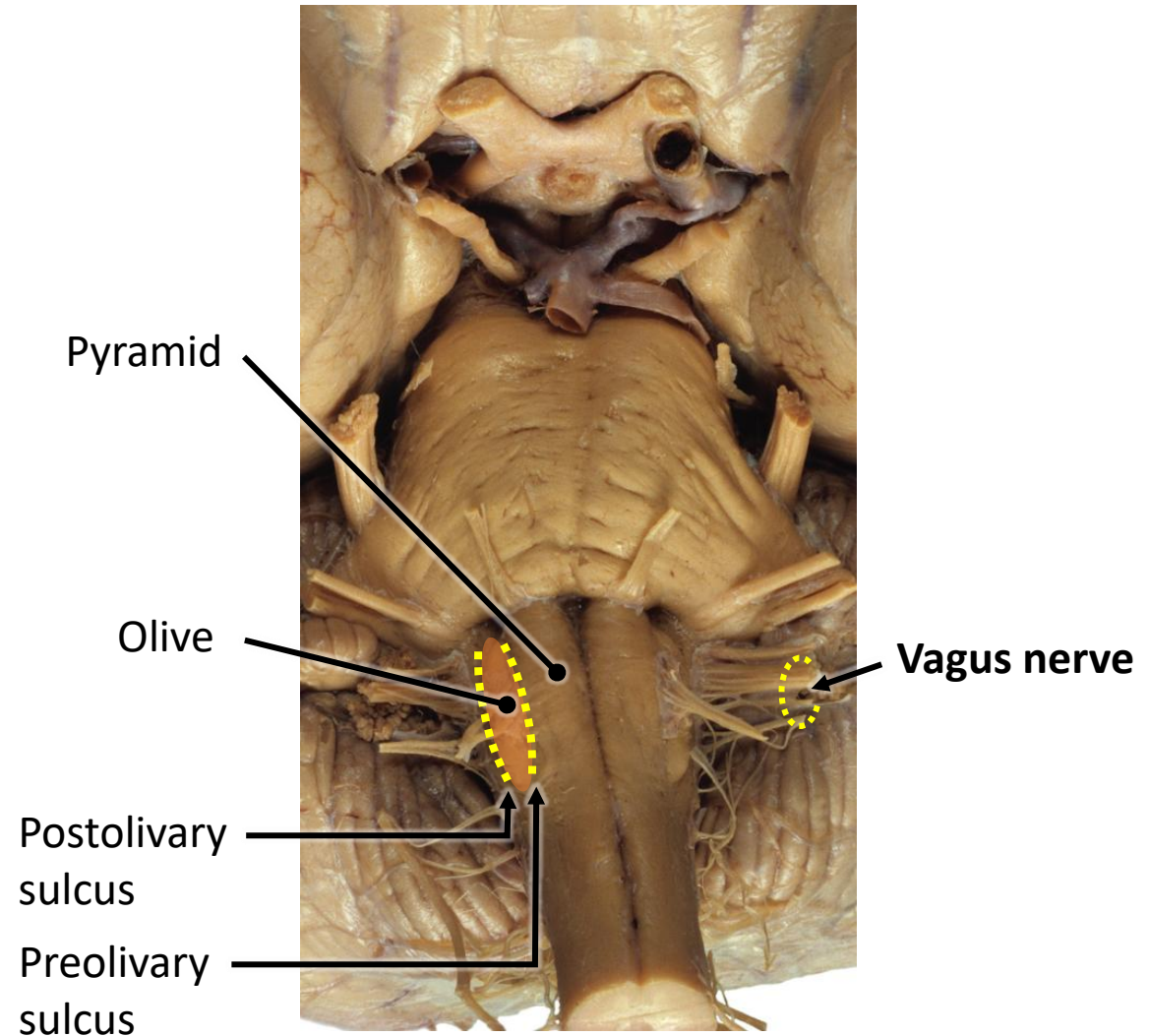
- Location:
 - Postolivary sulcus
- Function(s):
 - Somatic afferents: sensations of palate and posterior pharynx
 - Visceral afferents (special senses): gustation (taste) from posterior tongue
 - Visceral afferents: input from carotid sinus and carotid body (regulating cardiac function)
 - ...



Cranial Nerves of the Medulla (Anterior View)

- **CN X: Vagus nerve**

- Location:
 - Postolivary sulcus
- Function(s):
 - Somatic efferents: motor innervation of pharynx and larynx
 - Visceral afferents (special senses): gustation (taste) from base of tongue
 - ...



Cranial Nerves (Anterior View)

- **CN XI: Accessory nerve**

- Location:
 - Posterior to the medullary olive and lateral cervical spinal cord!
- Function(s):
 - Somatic efferents: motor innervation of trapezius (raising shoulders) and sternocleidomastoid muscles (head turning)

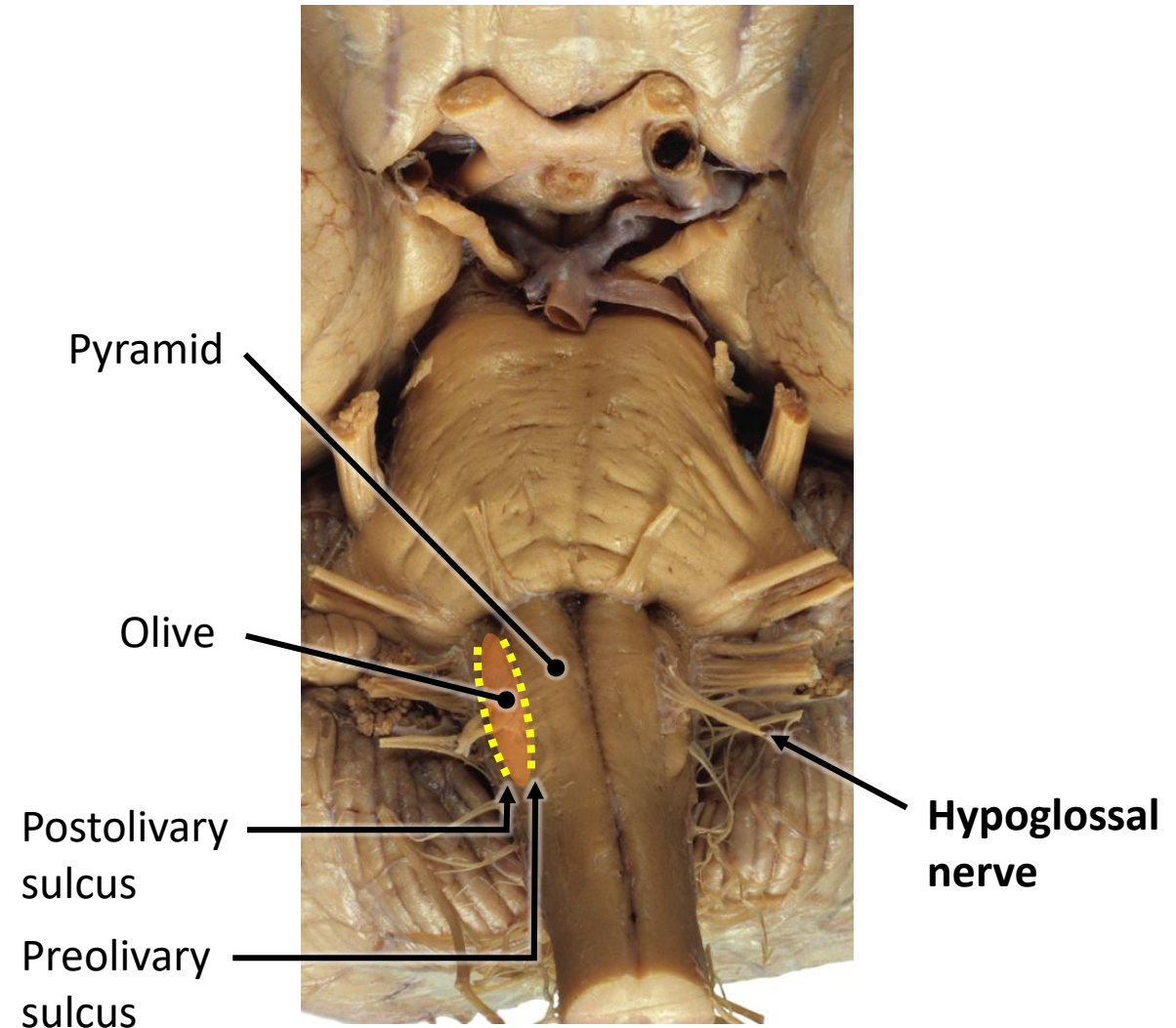
Accessory nerve



Cranial Nerves of the Medulla (Anterior View)

- **CN XII: Hypoglossal nerve**

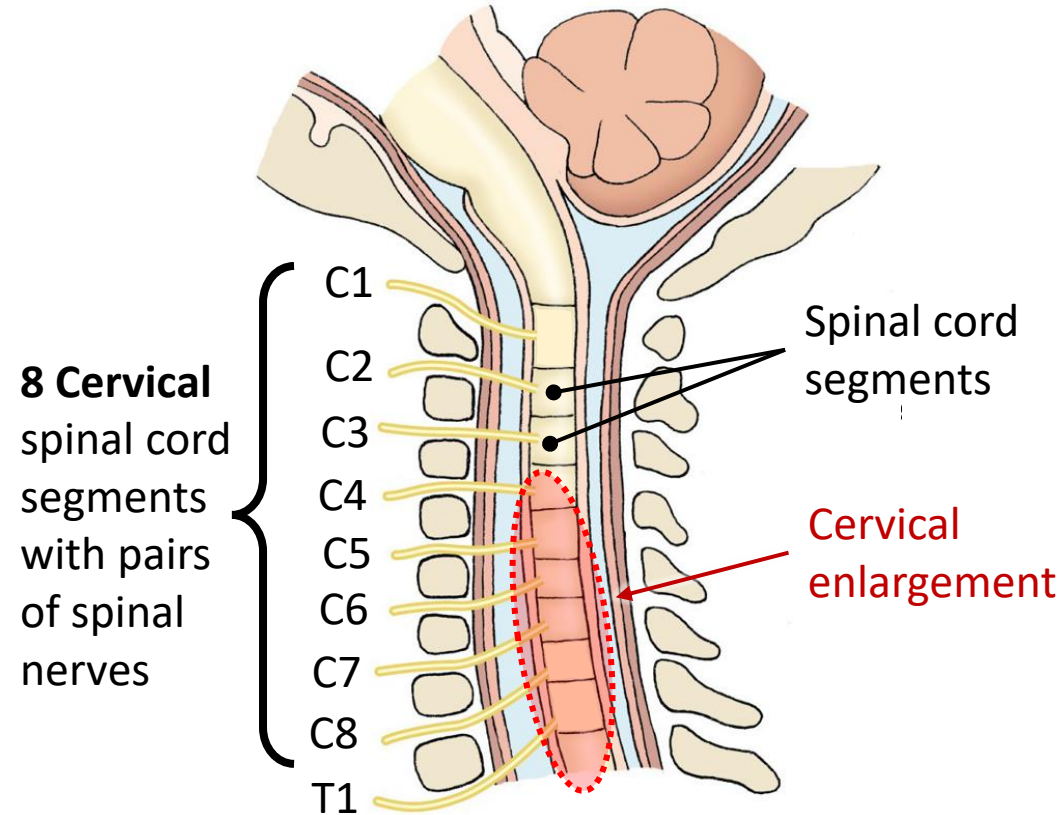
- Location:
 - Preolivary sulcus
- Function(s):
 - Somatic efferents: motor innervation of the tongue (including protrusion)



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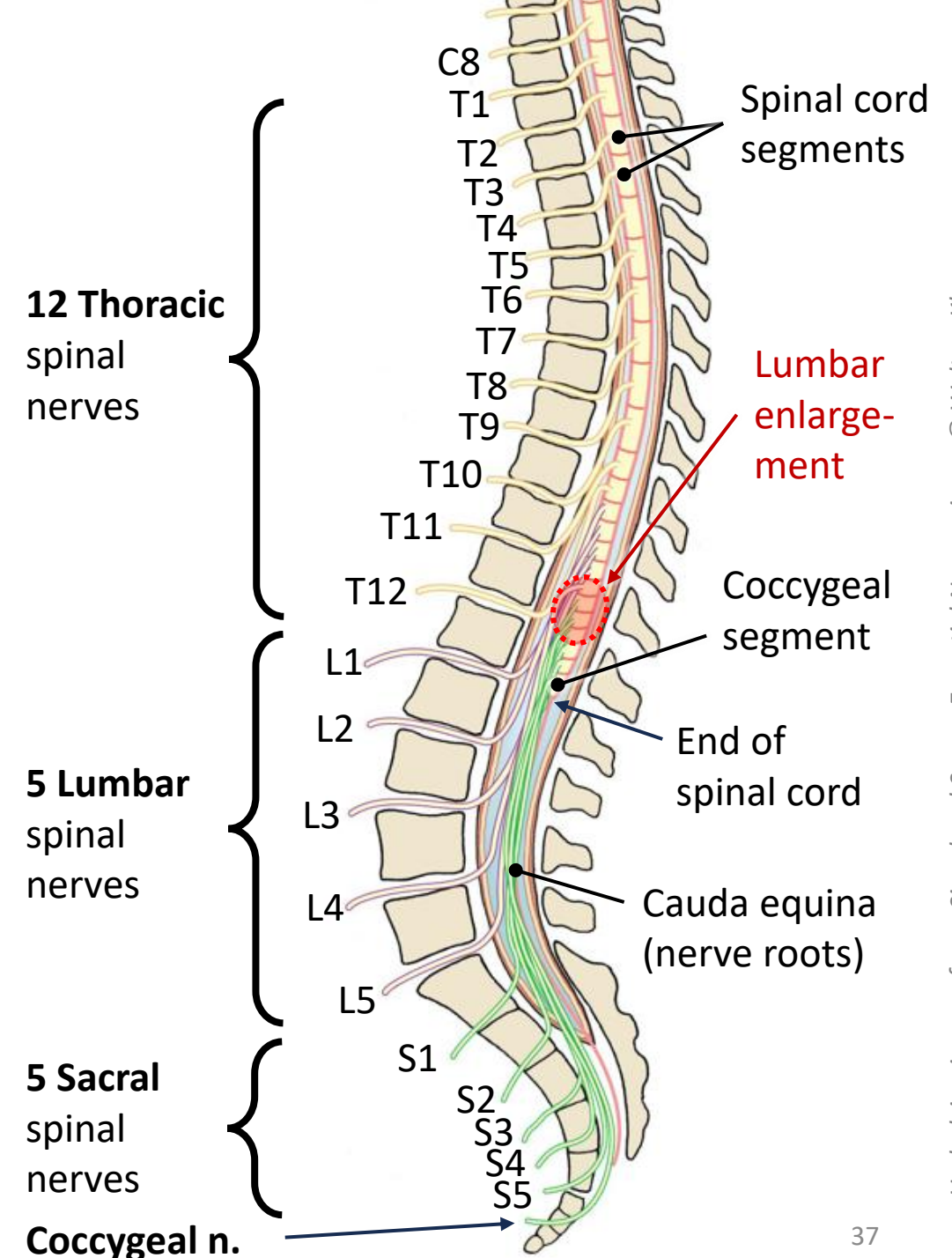
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Spinal Cord – Lateral View



- Upper extremities are innervated by C4 to T1 spinal cord segments (cervical enlargement)
- Lower extremities are innervated by L1 to S3 spinal cord segments (lumbar enlargement)

NSCI.0004



Significance of Cervical and Lumbar Enlargements

- The **cervical enlargement** of the spinal cord is a prominent macroscopic expansion of lower cervical (and upper thoracic) spinal cord segments. The enlargement is due to the increased number of neurons in control of complex movements of the upper extremities, including fingers, hands and arms, allowing us sophisticated manipulation of objects we are holding in our hands.
- The **lumbar enlargement** of the spinal cord is a prominent macroscopic expansion of lumbar (and upper sacral) spinal cord segments. The enlargement is due to the increased number of neurons in control of the powerful musculature of the lower extremities, allowing us to walk, run and jump.

Learning Objectives

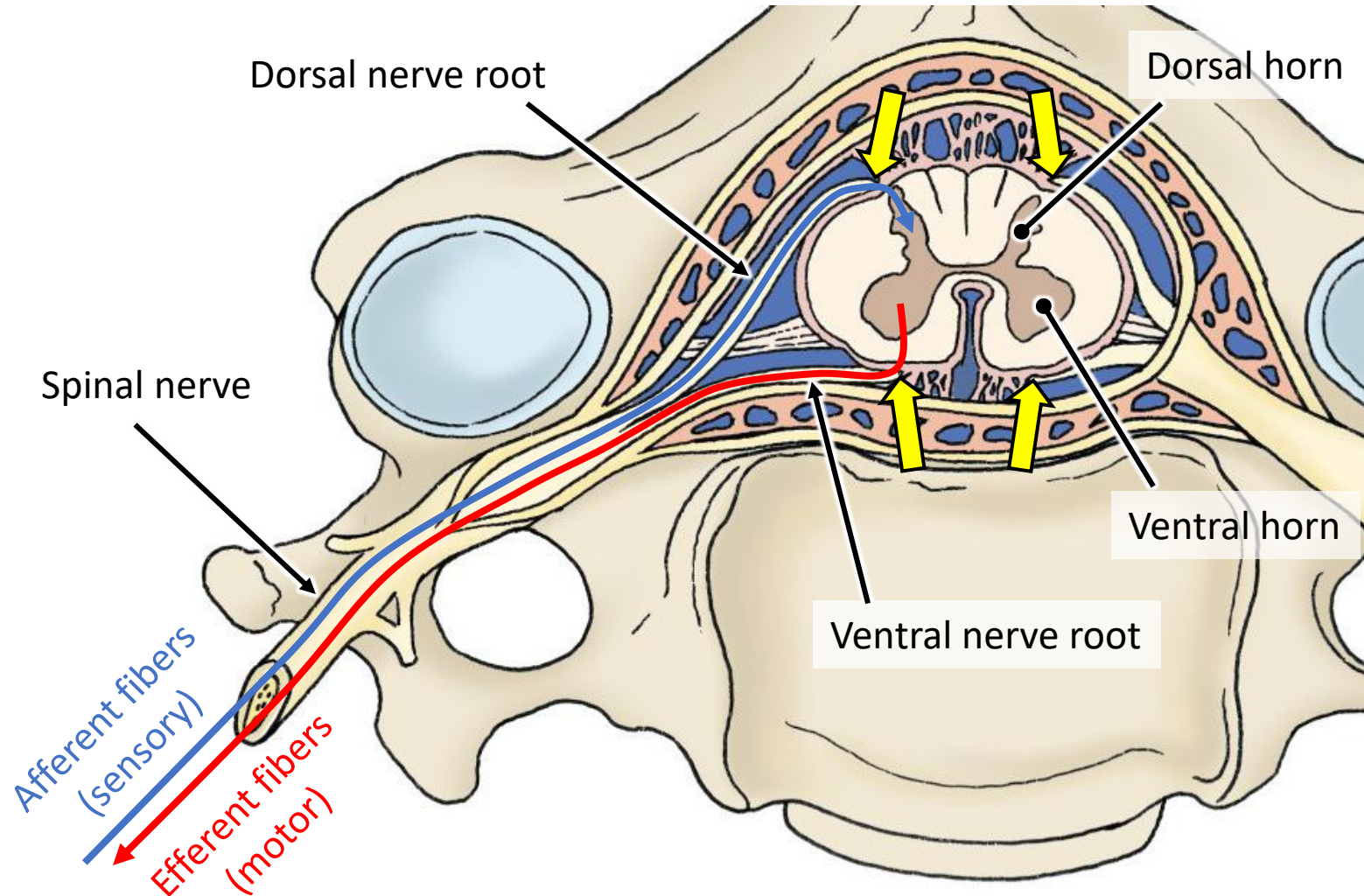
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Transition between Central and Peripheral Nervous System

- The transition zone between the Central Nervous System (CNS) and the Peripheral Nervous System (PNS) is known as **Redlich–Obersteiner zone**.
- It is located at the junction between cranial nerve roots and the brainstem, or between spinal nerve roots and the spinal cord.
- In this transition zone, myelination of axons changes from **oligodendrocytes** (the myelinating cells of the CNS) to **Schwann cells** (the myelinating cells of the PNS).
- **Note:** Some demyelinating diseases selectively affect oligodendrocytes (Multiple Sclerosis), and hence the CNS fibers, while other demyelinating diseases selectively affect Schwann cells (Guillain-Barré syndrome), and hence the PNS.

Transition between Central and Peripheral Nervous System

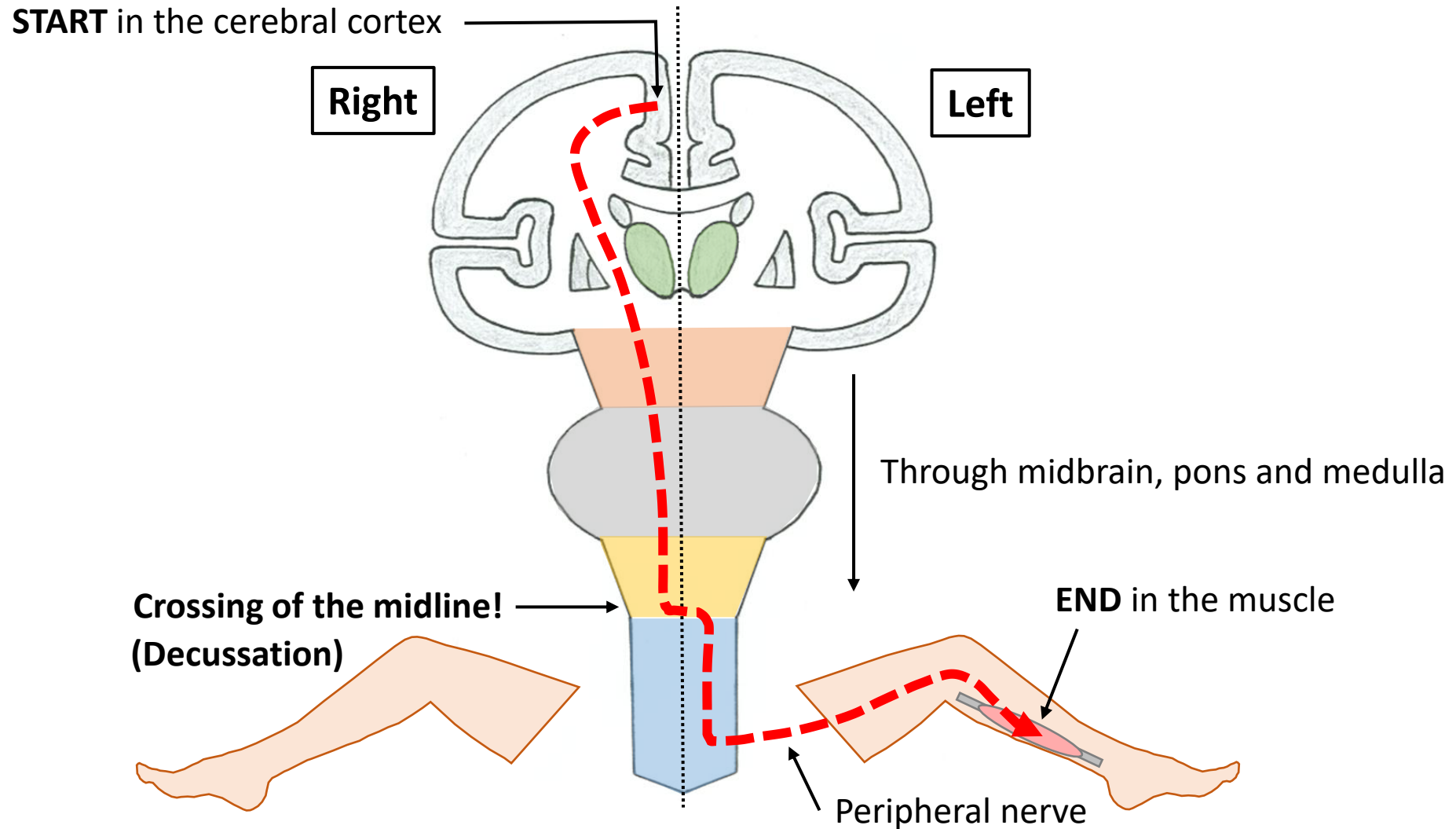
- Yellow arrows mark the **Redlich-Obersteiner zone**, the root entry zone of dorsal and ventral nerve roots, which is the transition zone between the central nervous system (CNS) and the peripheral nervous system (PNS)



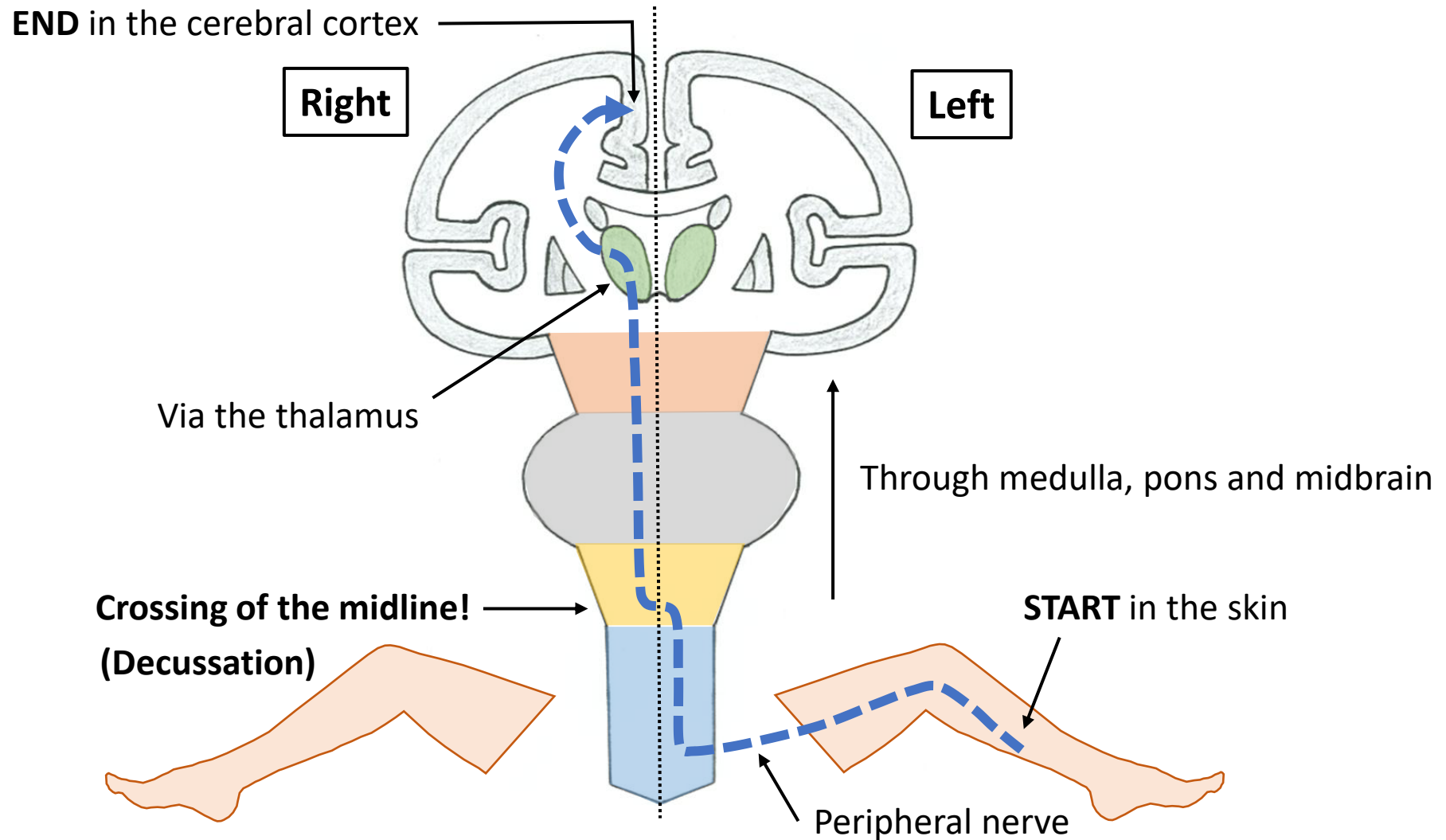
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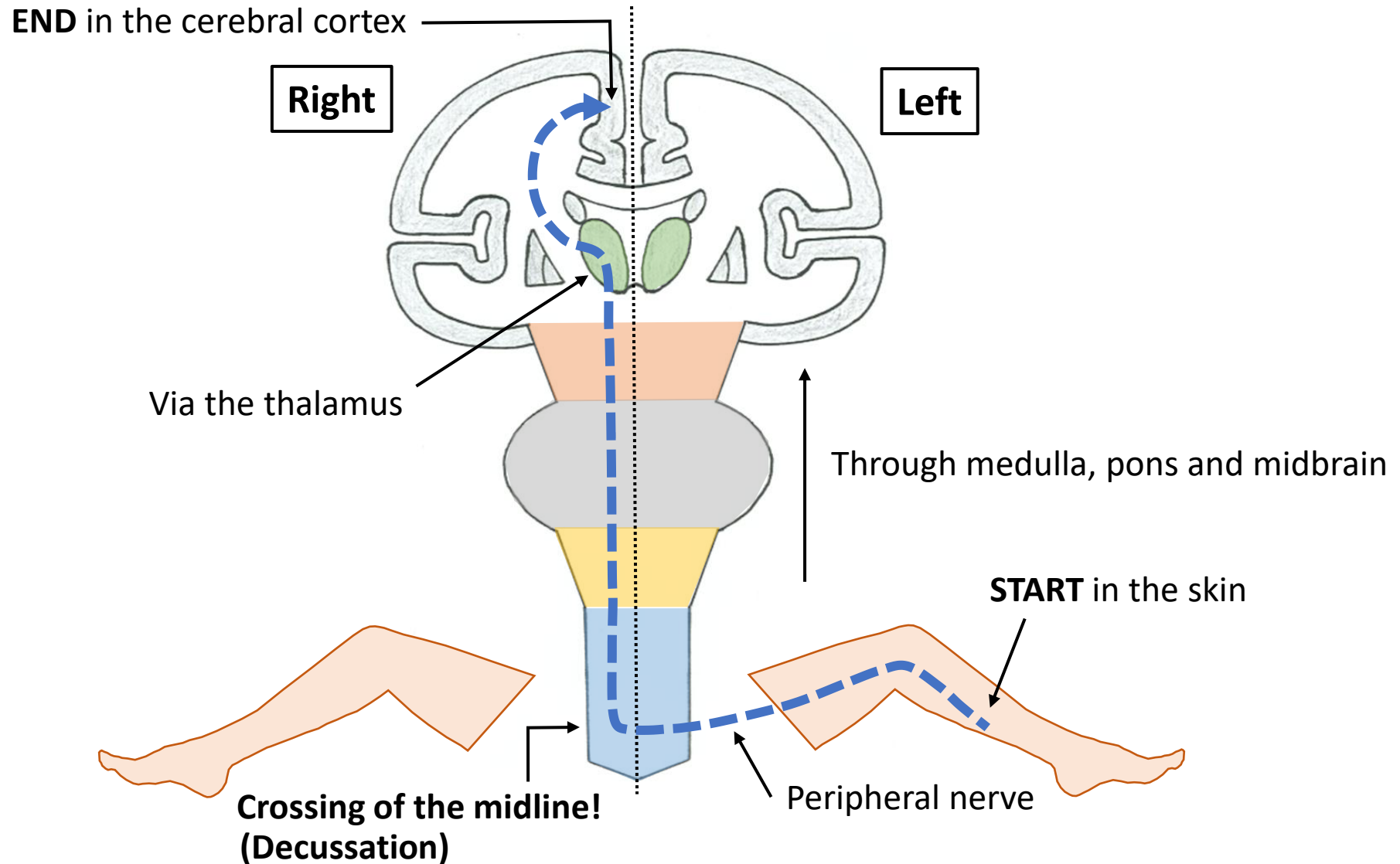
Communications between CNS and PNS – Motor Pathway



Communications between CNS and PNS – Touch Pathway



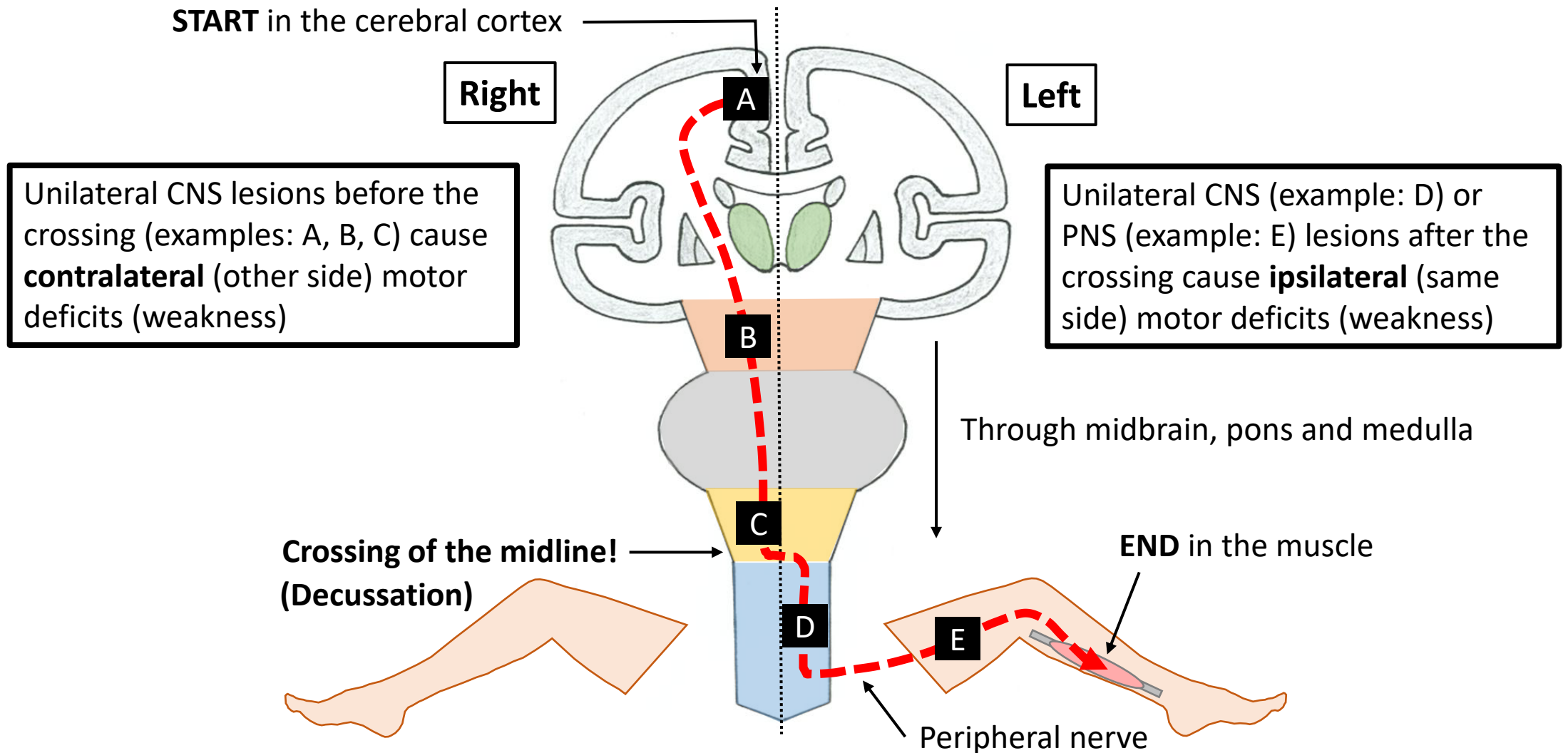
Communications between CNS and PNS – Pain Pathway



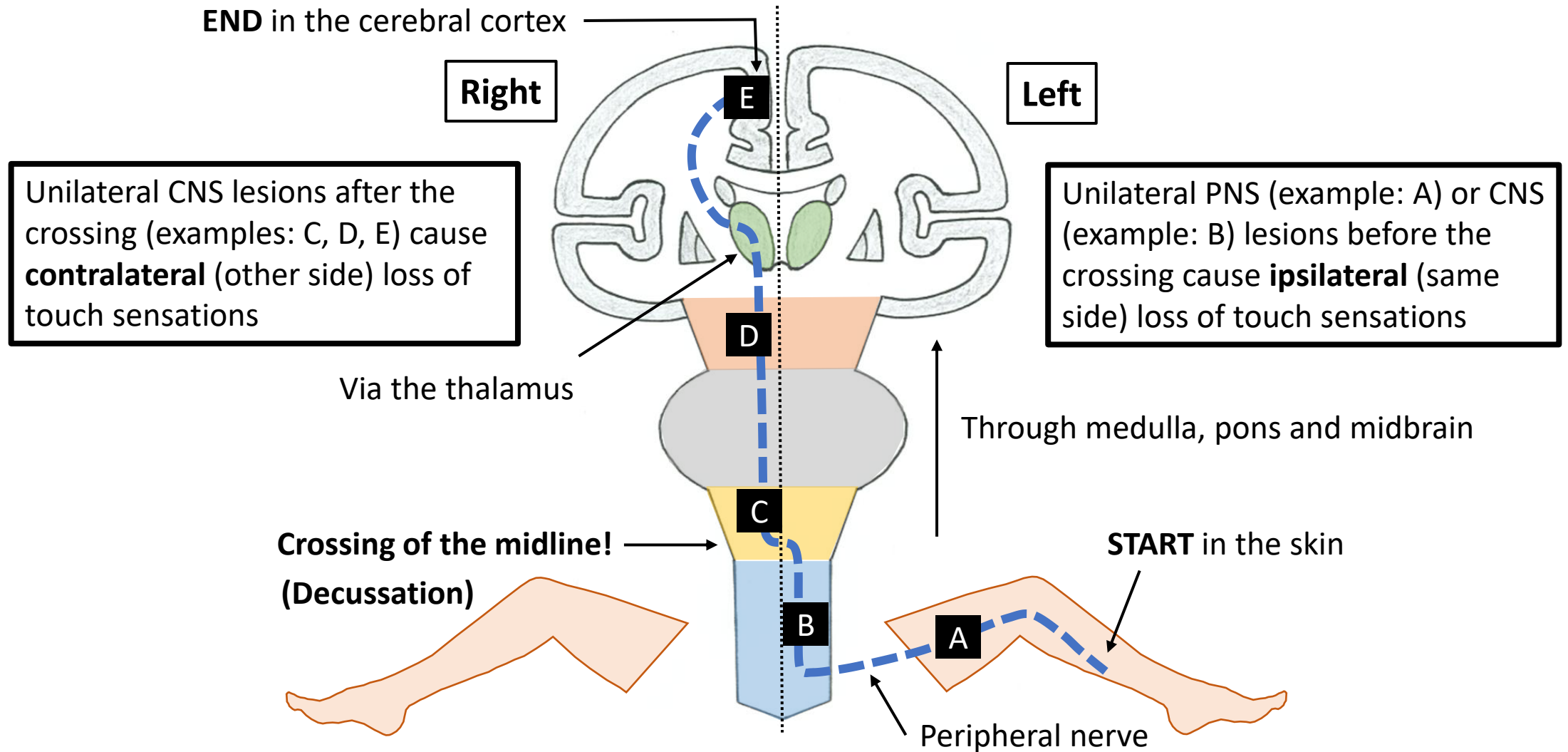
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- **SOM.MK.I.BPM2.1.NB.1.NSCI.0553** Describe the basic principles of somatic motor and sensory pathways
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0554** Explain how unilateral CNS or PNS lesions cause ipsilateral or contralateral functional deficits

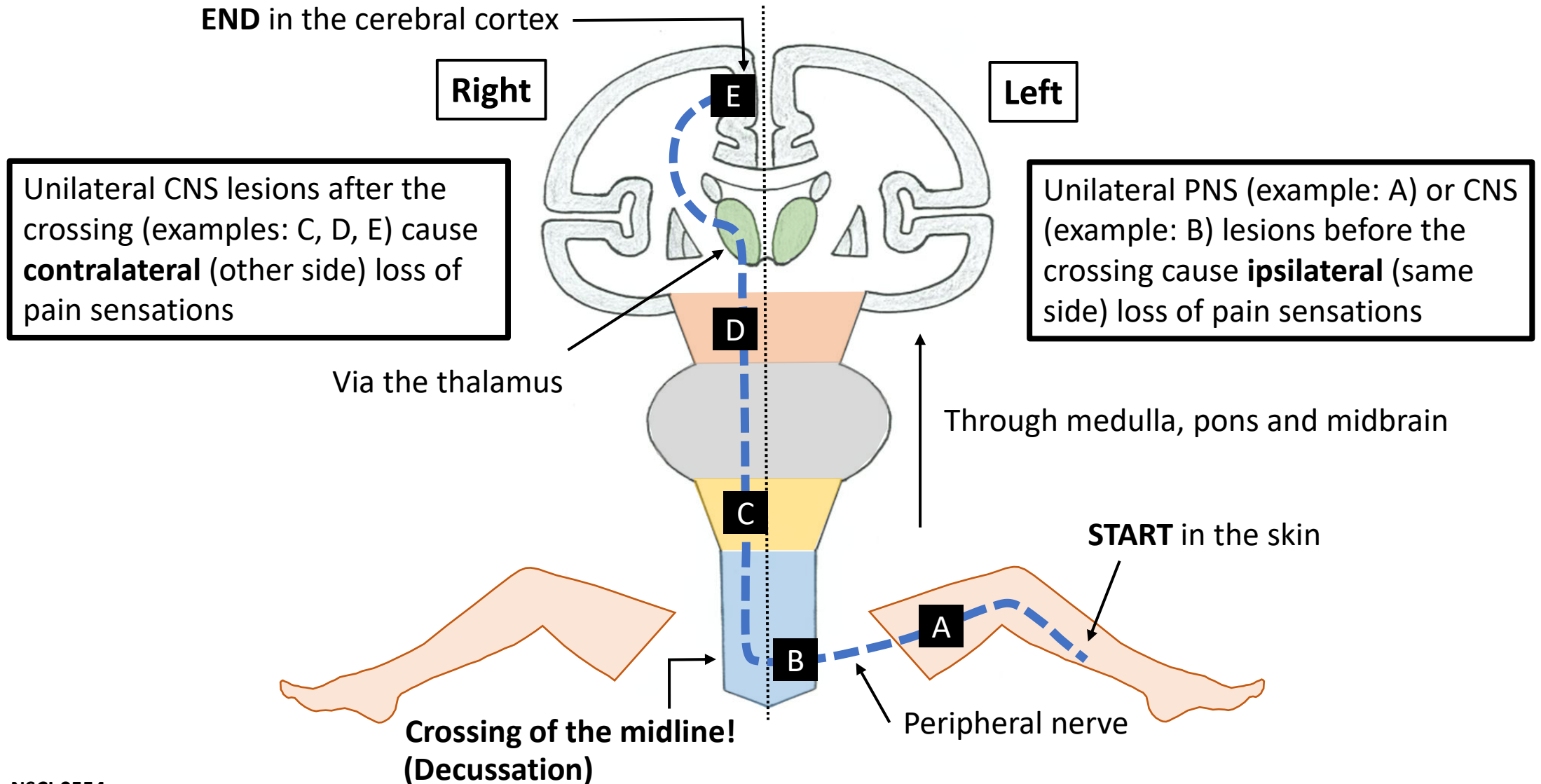
Motor Pathway Lesions



Touch Pathway Lesions



Pain Pathway Lesions



The End of Organisation of the Nervous System

If you have any questions, please send an email to:

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